

Identification of the lipoxygenase gene family from *Vitis vinifera* and biochemical characterisation of two 13-lipoxygenases expressed in grape berries of Sauvignon Blanc

Andriy Podolyan^A, Jackie White^A, Brian Jordan^A and Chris Winefield^{A,B}

^ADepartment of Wine, Food and Molecular Biosciences, Faculty of Agriculture and Life Sciences, Lincoln University, PO Box 84, Lincoln 7647, Canterbury, New Zealand.

^BCorresponding author. Email: chris.winefield@lincoln.ac.nz

Abstract. Lipoxygenases (LOXs) are a group of non-haem iron-containing dioxygenases that catalyse oxygenation of polyunsaturated fatty acids (PUFAs) and lipids, and initiate the formation of biologically active compounds known as oxylipins. Several plant oxylipins comprise important flavours and aromas in food and beverages. Analysis of the grape (*Vitis vinifera* L.) genome revealed that the grape LOX family consists of 18 individual members. Phylogenetic analysis places all except one of the identified grape LOXs into either a type II 13-LOX cluster or the type I 9-LOX cluster. Four LOX genes (*VvLOXA*, *VvLOXO*, *VvLOXC*, *VvLOXD*), representative of the major LOX groupings observed in the phylogenetic analyses, were selected for analysis of patterns of transcript abundance in berry tissues. *VvLOXA* and *VvLOXO* represent putative 13-LOXs, while *VvLOXC* represents a putative 9-LOX. *VvLOXD* represents a unique LOX that differs significantly from other characterised plant LOXs. All four LOXs exhibited a complex pattern of gene expression. Across all developmental stages, *VvLOXA* was the most abundant LOX and was expressed predominantly in berry skins. The expression pattern of *VvLOXC* and *-D* are more evenly distributed between seeds, pulp and skin, while *VvLOXO* is mostly expressed in the seed. Mechanical wounding and infection of berries with *Botrytis cinerea* Pers.: Fr resulted in rapid accumulation of *VvLOXC* and *-O* transcripts. *VvLOXA* expression decreased in diseased berries. Biochemical analysis of *VvLOXA* and *-O* recombinant proteins confirmed that these LOX genes encode functional 13-LOXs that exhibit different pH and temperature optima. Both enzymes showed activity with linoleic, linolenic and arachidonic acids.

Additional keywords: berry development, grapevine, green leaf volatiles, LOX, oxylipins, plant–pathogen interaction.

Introduction

Lipoxygenases (LOX; EC 1.13.11.12) are a group of enzymes that initiate the degradation of free fatty acids and esterified lipids via various branches of the LOX pathway (Fig. 1). LOXs are non-haem iron-containing dioxygenases that oxygenate polyunsaturated fatty acids (PUFAs) containing a *cis,cis*-1,4-pentadiene structural unit to produce either 13(S)- or 9(S)- fatty acid hydroperoxides (henceforth termed hydroperoxides). Hydroperoxides generated by LOX action are rapidly converted either chemically or enzymatically into a range of biologically active compounds known as oxylipins (Fig. 1). Hierarchical organisation of the pathway provides metabolic branch-points at either LOX action (13- v. 9-LOX activity) or CYP74 subfamily action. The CYP74 enzyme family are a grouping of atypical cytochrome P450 monooxygenases that include allene oxide synthase (AOS), hydroperoxide lyase (HPL), divinyl ether synthase (DES) and epoxy alcohol synthase (EAS) (Fig. 1) (Stumpe and Feussner 2006). In plants, oxylipins have been implicated in a wide range of important physiological functions such as hormone signalling, defence against pathogens and herbivores, abiotic stress response, growth, development and senescence (reviewed by Feussner and Wasternack 2002; Porta

and Rocha-Sosa 2002; Liavonchanka and Feussner 2006; Wasternack 2007). Oxylipins are also commercially important compounds, as many are utilised as components of fragrances and contribute to flavours and aromas of food and beverages (Casey *et al.* 1999; Schwab *et al.* 2008).

In grapes (*Vitis vinifera* L.), oxylipins such as jasmonic acid (JA) and its derivatives have been proposed to be involved in grape development as well as in defence against pests and pathogens (reviewed by Peña-Cortés *et al.* 2004). To date, however, the best studied grape oxylipins are the C6 volatile aldehydes, alcohols and esters that are formed through the combined LOX and HPL mediated metabolism of free fatty acids. C6 volatiles are collectively known as green leaf volatiles (GLVs) because of their characteristic aroma that is often associated with the scent of freshly cut grass (Matsui 2006). It has been observed that the content and composition of different forms of GLVs change during grape berry development and in response to postharvest water stress (Chkaiban *et al.* 2007; Kalua and Boss 2009).

C6 compounds also form rapidly upon berry crushing and during maceration of grape must, and have been reported to constitute 93% of the total volatile compounds in Chardonnay

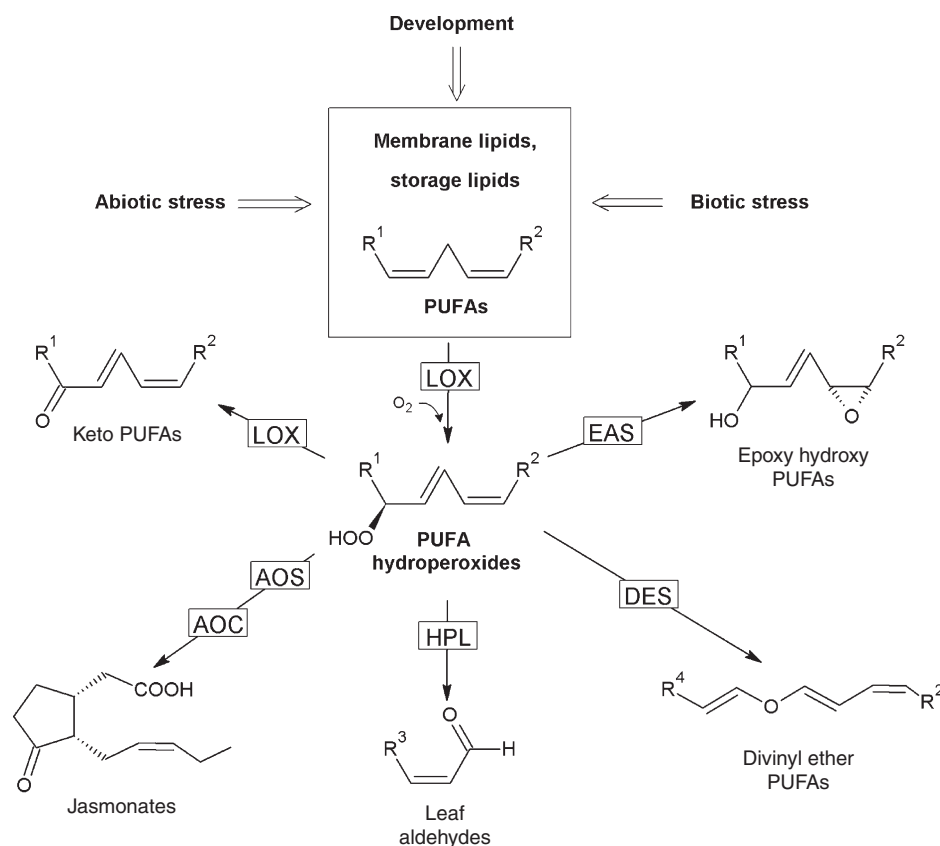


Fig. 1. LOX pathway. Metabolism of the PUFA-containing lipids is initiated by the enzymatic action of LOXs. LOXs are regulated by various developmental and environmental stimuli. The end-products of LOX action are fatty acid hydroperoxides. Hydroperoxides can be further metabolised either by LOX itself or by members of the CYP74 family of atypical cytochrome p450s. CYP74 action leads to the production of oxylipins. AOS, allene oxide synthase; AOC, allene oxide cyclase; HPL, hydroperoxide lyase; DES, divinyl ether synthase; EAS, epoxy alcohol synthase.

grapes and musts (Ramey *et al.* 1986; Baumes *et al.* 1988). The presence of these compounds in wine can either be beneficial (bringing fresh, herbaceous notes to white varieties) or undesirable (bringing unripe characters to red varieties) (Clarke and Bakker 2004). C6 compounds have also been implicated as precursors of important volatile thiols in wine (Schneider *et al.* 2006). Volatile thiols such as 3-mercaptohexyl acetate (3MHA) and 3-mercaptohexan-1-ol (3MH) were found to be amongst the most important aroma compounds determining the unique character of New Zealand Marlborough Sauvignon Blanc wine (Lund *et al.* 2009).

There is relatively little known about grape lipoxygenases. A single putative LOX-gene sequence has been used in transcriptome-wide gene expression (da Silva *et al.* 2005) and also included in several grape defence response studies (Chong *et al.* 2008; Petit *et al.* 2009). However, to the best of our knowledge, there has been no extensive molecular or biochemical characterisation of the grape LOX family reported to date.

In this paper, we have attempted to determine the size and structure of the grape LOX family. Additionally, we sought to provide an initial molecular and biochemical characterisation of

individual LOXs that represent members of the major grape LOX groupings, focusing on those expressed in grape berries and on biochemical analysis of those LOXs that are tentatively involved in C6 compound formation. The results of this and subsequent studies will help to increase our understanding of the roles of lipoxygenases in grape development as well as stress and defence response mechanisms in grapevines. This knowledge will also have a practical application in establishing links between viticultural practice, changes in grapevine physiology, and their combined influence on grape and wine quality.

Materials and methods

Plant material and field trial conditions

Developmental stages

The plant material utilised in this study was harvested from commercially grown vines that consisted of the mass selected clone UCD1 of the white grape cultivar Sauvignon Blanc (*Vitis vinifera* L., cv. Sauvignon Blanc) grafted onto SO4 rootstock. Berries for gene expression studies were collected at various points during berry development in the 2007, 2008 and 2009 seasons from the Booker vineyard (Montana™ New Zealand

Brancott estate), Marlborough, New Zealand. Each season, samples were collected at eight berry developmental stages. Initial berry sampling took place 20 days after anthesis (daa). Subsequent samplings were undertaken at the following times: 30 daa, 50 daa, 60 daa, 80 daa, 90 daa, 100 daa and 110 daa. In each season, veraison took place at ~50–60 daa. Berries harvested at 110 daa had soluble solid levels of between 21–23° Brix and were considered fully ripe.

Developmental variation in berries within a bunch, between bunches on a vine and between bunches on different vines represents a significant issue for sample collection in grapevines. A recent study has shown that individual berries ‘calibrated’ for developmental stage (for example on the basis of Brix, texture or colour) are remarkably coordinated in their transcriptional activity, suggesting that pools of such berries are less likely to be highly variable than berries taken at random from individual bunches (Lund *et al.* 2008). Therefore, in an attempt to reduce berry to berry variation in our sampling, bunches from the basal position of 20–30 independent vines were harvested and only those berries typical for a given stage were collected. Harvested berries were immediately frozen in liquid nitrogen. The pool of berries for each stage was collected at random from 20–30 different vines. Harvested material was stored at –80°C until use.

Berry fractions

In order to investigate the tissue-specific gene expression patterns of selected LOXs that may contribute to the formation of short chain aliphatic compounds in juice for wine making Sauvignon Blanc berries of between 80 and 90 daa were collected in the 2009 season from the Lincoln University research vineyard in Canterbury, New Zealand. This material was used to analyse the distribution of transcripts between skin, pulp and seed fractions of berries. The skin fraction was separated by squeezing the pulp out of the berry. The remaining part of the berry was then pressed through a sieve to separate the seed fraction and the pulp fraction. All three fractions were frozen immediately and stored in a freezer at –80°C until use.

Wounding

Wounding experiments were performed in the 2008 season using material harvested from the Lincoln University research vineyard, Canterbury, New Zealand on Sauvignon Blanc berries at 80–90 daa. Grape berry clusters were maintained on the vine and pierced with a 12 gauge needle so that most berries in a cluster were pierced 1–2 times. Three independent wounded clusters from different vines were then collected at each of the following time intervals after wounding: 0.5 h, 1 h, 3 h, 6 h, 9 h, 12 h and 24 h. The clusters were snap-frozen in liquid nitrogen and stored at –80°C before further use. Three clusters on vines separate to those used in the wounding trial were collected before wounding for use as an untreated control.

Pathogen infected berries

Plant material for studying LOX gene expression in berries infected with the grape pathogen *Botrytis cinerea* Pers.: Fr was collected in the 2008 season from the Booker vineyard, Marlborough, New Zealand, at harvest (between 100 and 110 daa). Three groups of samples were collected: berries from

healthy bunches, berries with visible signs of pathogen infection and berries from the infected bunches without visible signs of pathogen infection. The berries were snap-frozen in liquid nitrogen and stored in a freezer at –80°C until use.

In silico mining of the grape genome for the LOX gene family

The grape genome (*V. vinifera*, cv. Pinot Noir, quasi-homozygous genotype PN40024) sequence was analysed using a set of web-based tools provided by the French-Italian Public Consortium (<http://www.genoscope.cns.fr/spip/Vitis-vinifera-e.html>, accessed 21 May 2010). Sequences of known plant LOXs were used in BLAST and BLAT alignments in order to find similar sequences in the grape genome. Identified genomic sequences were then analysed to determine their intron–exon structure and predicted coding sequence regions using tools provided by GenScan (<http://genes.mit.edu/GENSCAN.html>, accessed 21 May 2010). The location of identified LOX sequences on individual grape chromosomes or linkage groups was determined using the Grape Genome Browser <http://www.genoscope.cns.fr/externe/GenomeBrowser/Vitis/>, accessed 21 May 2010). The corresponding nucleotide and protein accession numbers for some of the identified LOX genes were retrieved from NCBI GenBank (<http://www.ncbi.nlm.nih.gov/Genbank/index.html>, accessed 21 May 2010).

Phylogenetic tree construction

DNA and amino acid sequences of the predicted grape LOXs and several biochemically characterised plant LOXs were retrieved from GenBank or UniProt. Neither database searches nor 5′RACE experiments were able to retrieve a putative full-length clone for *VvLOXD*. For the phylogenetic tree construction, the largest fragment of *VvLOXD* retrieved from 5′RACE experiments was therefore used. Amino acid sequences were aligned using ClustalW (Thompson *et al.* 1994) using the MEGA 4.0 software package (Tamura *et al.* 2007). The same software package was used to generate an unrooted phylogenetic tree based on the alignments obtained from the Neighbour–Joining method (Saitou and Nei 1987). To validate the results obtained from an alignment of full-length sequences, new alignments and phylogenetic tree reconstructions were generated from sequences containing only the LOX domain. The sequences of the characterised plant LOXs used in the alignment were named according to Feussner and Wasternack (2002) and to proposed nomenclature protocols (Shibata *et al.* 1994). Nucleotide and protein accession numbers for the sequences used in the alignment are included in Tables 1 and 2.

PCR and cloning

RNA isolation

RNA samples were extracted from a pool of 3–4 randomly selected Sauvignon Blanc grape berries. In order to minimise potential berry to berry variation, the pool for each stage represented berries collected from 20–30 different vines, collected as described previously. Berries or their individual fractions (skins, pulp and seeds) were ground to a fine powder under liquid nitrogen using a mortar and pestle. Total RNA was extracted from 200 mg of ground berry tissue (100 mg for seeds)

Table 1. Identified grape LOXs and predicted size of coding sequence (CDS), chromosome location, NCBI accession and proposed functionality

ID	NCBI nucleotide accession	Predicted CDS (bp)	Chromosome (strand)	NCBI protein accession	Predicted protein length (aa)	Predicted functionality
<i>VvLOXA</i>	XM_002285538	2706	Chr 6 (–)	XP_002285574	901	13-LOX
<i>VvLOXB</i>	XM_002280615	2601	Chr 14 (–)	XP_002280651	866	9-LOX
<i>VvLOXC</i>	XM_002284499	2580	Chr 14 (–)	XP_002284535	859	9-LOX
<i>VvLOXD</i>	N/A	2177	Chr Un Random (+)	N/A	725	Unknown
<i>VvLOXE</i>	XM_002283130	2718	Chr 6 (+)	XP_002283166	905	13-LOX
<i>VvLOXF</i>	XR_078140	2607	Chr 6 (+)	N/A	869	Pseudogene
<i>VvLOXG</i>	XM_002283111	2712	Chr 6 (+)	XP_002283147	903	13-LOX
<i>VvLOXH</i>	XM_002283099	2610	Chr 6 (+)	XP_002283135	869	13-LOX
<i>VvLOXI</i>	XM_002283087	2511	Chr 6 (+)	XP_002283123	836	13-LOX
<i>VvLOXJ</i>	XM_002263818	2712	Chr 13 (+)	XP_002263854	903	13-LOX
<i>VvLOXK</i>	XR_077340	2208	Chr 13 (+)	N/A	736	Pseudogene
<i>VvLOXL</i>	XM_002277971	2631	Chr 5 (+)	XP_002278007	876	9-LOX
<i>VvLOXM</i>	N/A	1062	Chr 6 (–)	N/A	353	Pseudogene
<i>VvLOXN</i>	N/A	267	Chr 14 (–)	N/A	89	Pseudogene
<i>VvLOXO</i>	XM_002273222	2751	Chr 9 (+)	XP_002273258	916	13-LOX
<i>VvLOXP</i>	XM_002265469	2763	Chr 1 Random (+)	XP_002265505	920	13-LOX
<i>VvLOXR</i>	XM_002265785	1847	Chr Un-random (–)	XP_002265821	614	Pseudogene
<i>VvLOXS</i>	XM_002275232	2760	Chr Un-random (–)	XP_002275268	919	13-LOX

Table 2. Functionally characterised plant LOXs used in multiple alignments and phylogenetic tree construction

Species	ID	NCBI nucleotide accession	UniProt or NCBI protein accession
<i>Arabidopsis thaliana</i>	<i>LOX1_At_1</i>	AY093104	Q06327
	<i>LOX1_At_2</i>	AJ302043	Q9LUW0
	<i>LOX2_At_1</i>	AY062611	P38418
	<i>LOX2_At_2</i>	AY075625	Q9LNR3
	<i>LOX2_At_3</i>	AY056166	Q9FNX8
	<i>LOX2_At_4</i>	AC011020; F12B7.11	Q9CAG3
<i>Arachis hypogaea</i> L.	<i>LOX1_Ah_1</i>	AF231454	Q9M5D3
<i>Brassica napus</i> L.	<i>LOX2_Bn_2</i>	AY162143	Q8GV01
<i>Glycine max</i>	<i>LOX1_Gm_1</i>	J02795	P08170
	<i>LOX1_Gm_2</i>	J03211	Q39870
	<i>LOX1_Gm_3</i>	X13302	P09186
	<i>LOX1_Gm_4</i>	D13999	P38417
	<i>LOX1_Gm_5</i>	U50075	Q43446
	<i>LOX1_Gm_6</i>	U26457	Q43440
	<i>LOX1_Gm_7</i>	X56139	P24095
<i>Hordeum vulgare</i> L.	<i>LOX1_Hv_1</i>	L35931	P29114
	<i>LOX2_Hv_1</i>	U56406	P93184
<i>Lens culinaris</i> L.	<i>LOX1_Lc_1</i>	X71344	P38414
<i>Nicotiana tabacum</i> L.	<i>LOX1_Nt_1</i>	X84040	Q43800
<i>Oryza sativa</i> L.	<i>LOX1_Os_1</i>	X64396	P29250
	<i>LOX2_Os_1</i>	D14000	P38419
	<i>LOX2_Os_2</i>	AJ270938	Q9FSE5
<i>Pisum sativum</i> L.	<i>LOX1_Ps_2</i>	X78580	P14856
	<i>LOX1_Ps_3</i>	X78581	P09918
	<i>LOX2_Pod_1</i>	DQ131178	Q45HK7
<i>Populus deltoids</i> (Bartr. Ex Marsh.)	<i>LOX2_Pod_2</i>	DQ131179	Q45HK6
	<i>PtLOX</i>	XM_002320535	XP_002320037
<i>Populus trichocarpa</i> (Torr. & Gray)	<i>LOX1_Prd_1</i>	AJ418043	Q8W4X6
<i>Prunus dulcis</i> (Mill.)	<i>LOX1_St_1</i>	X79107	P37831
	<i>LOX1_St_2</i>	U24232	O49150
	<i>LOX2_St_1</i>	X96405	O24370
	<i>LOX2_St_2</i>	X96406	O24371
	<i>LOX1_Zm_1</i>	DQ335760	Q9 LKL4
<i>Zea mays</i>	<i>LOX1_Zm_3</i>	AF329371	Q9AXG8
	<i>LOX2_Zm_1</i>	DQ335768	A1XC15
	<i>LOX2_Zm_6</i>	NM_001112506	A1XC11

using Spectrum™ Plant Total RNA Kit (Sigma Aldrich, Auckland, New Zealand) as per the manufacturer's instructions. The eluted RNA samples were treated with TURBO DNA-free™ Kit (Ambion, Geneworks, Auckland, New Zealand) to remove traces of genomic DNA. RNA was then precipitated using a 1-in-10 sample volume of 3 M sodium acetate, 2 volumes of ethanol, washed with 70% ethanol and resuspended in 30 µL of nuclease-free water. The concentration and quality of RNA was determined by measurement of 260:280 nm and 260:230 nm ratios of the purified RNA, estimated using a NanoDrop® ND-1000 Spectrophotometer (Thermo Scientific, Auckland, New Zealand). The integrity of the extracted RNA was determined visually using denaturing agarose gel electrophoresis (Sambrook and Russell 2001).

Complementary DNA preparation

Complementary DNA was synthesised from total grape RNA using an Oligo-dT primer and PrimeScript™ Reverse Transcriptase (Takara Bio Inc., Norrie Biotech, Auckland, New Zealand) according to the manufacturer's recommendations. Up to 10 µg of total RNA were used in a 40 µL reverse transcriptase reaction. The reaction was incubated at 42°C for 60 min and heated at 70°C to deactivate the enzyme.

PCR and cloning

For the grape LOX genes *VvLOXA*, -C and -O, primers to amplify full-length coding sequences were designed based on the *in silico* prediction (Table S1, available as an Accessory publication to this paper). High fidelity PrimeSTAR™HS DNA Polymerase (Takara Bio Inc.) was used in PCR reactions according to the manufacturer's instructions. The PCR products obtained were cloned into a *pENTR™/TEV/D-TOPO*® vector (Invitrogen, Auckland, New Zealand) and transformed into One Shot® TOP10 competent *Escherichia coli* cells (Invitrogen). Plasmids from six randomly selected clones for each gene were extracted, checked by restriction digest and sequenced using an ABI3730 Genetic Analyzer from ABI Inc. (AWCGS, Palmerston North, New Zealand).

Construction of *VvLOXA* and *VvLOXO* expression vectors

In silico prediction of subcellular localisation, the presence or absence of signal peptides and their possible localisation was determined using the TargetP 1.1 (<http://www.cbs.dtu.dk/services/TargetP/>, accessed 21 May 2010) (Emanuelsson *et al.* 2007), ChloroP 1.1 (<http://www.cbs.dtu.dk/services/ChloroP/>, accessed 21 May 2010) (Emanuelsson *et al.* 1999) and WoLF PSORT (<http://wolfpsort.org/>, accessed 21 May 2010) servers (Horton *et al.* 2007).

Based on predictions generated from the *in silico* analyses, PCR primers were designed to amplify full-length or truncated sequences with putative targeting or signalling peptide sequences removed (Table S2). Primers *LOXAFL_TOPO* and *LOXATerm* were used to amplify the full-length *VvLOXA* coding sequences, and *LOXA-TP_TOPO* and *LOXATerm* were used to amplify coding sequences with the putative chloroplast targeting peptide removed. Similarly, the primers *LOXOFL_TOPO* and *LOXO-TP_TOPO* were used in combination with the *LOXOTerm*

primer to amplify full-length and truncated (without putative targeting peptide) *VvLOXO* coding sequences respectively.

Full-length clones of *VvLOXA* (FJ858255) and *VvLOXO* (FJ858257) vectors were used as templates to produce expression constructs. Typical PCR reactions were performed with high-fidelity PrimeSTAR™HS DNA Polymerase (Takara Bio Inc.) according to the manufacturer's instructions with the following PCR conditions: 30 cycles (denaturation at 98°C for 10 s, annealing at 50°C for 15 s, extension at 72°C for 3.5 min, final extension at 72°C for 7 min). The PCR product was purified using "the AxyPrep kit (Axygen, Total Laboratory Systems, Auckland, New Zealand), cloned into a *pENTR™/TEV/D-TOPO*® vector (Invitrogen) and transformed into One Shot® TOP10 competent *E. coli* cells (Invitrogen). The resulting entry vectors were used to deliver the LOX constructs into the recombinant protein expression vector Gateway® *pDEST17* (Invitrogen) using Gateway® LR Clonase™ II Enzyme Mix (Invitrogen) according to the manufacturer's instructions. Reaction products were transformed into BL21(DE3) cells (Novagen, Merck Bioscience, Auckland, New Zealand). Plasmid DNA isolated from transformed BL21(DE3) cultures were extracted, checked by restriction digestion and sequenced as described before.

Semiquantitative real-time PCR

A Rotor-Gene™ 6000 real-time rotary analyser (Corbett Life Sciences) was used to conduct relative quantification of Sauvignon Blanc berry LOX transcripts. For the selected LOX genes, as well as four reference genes, PCR primers were designed using Vector NTI (Invitrogen) to unique regions within the 3'-end of their respective coding sequences (Table S3). The high levels of sequence similarity between *VvLOXA* and other closely related grape LOXs were taken into account when designing quantitative real-time PCR (qPCR) primers to avoid cross-annealing of primers to these sequences.

The cDNA templates used in this analysis were prepared and evaluated as described above. Serial dilutions of PCR products for each gene were used for generating standard curves. A 100-well gene-disk was used for each run, which contained samples consisting of serial dilutions for run-specific standard curves, and sample and non-template control reactions. A typical 10-µL qPCR reaction consisted of 5 µL of 2 × SYBR® Premix Ex Taq™ PCR Mix (Takara Bio Inc.), 1 µL of 2 µM primer mix (forward and reverse primers), 3 µL of diluted cDNA template and 1 µL of nuclease-free water. The PCR conditions were as follows: initial denaturation at 95°C for 15 s; 40 × 95°C for 5 s, 58°C for 15 s, 72°C for 15 s. Melt curve analysis was performed immediately after the amplification to confirm that the correct PCR products were formed. Reference genes used and their GenBank accession numbers were as follows: *Actin* (actin 1, XM_002282480), *GAPDH* (glyceraldehyde-3-phosphate dehydrogenase, XM_002263109), *SAND* (SAND family protein, XM_002285134) and *EF1* (elongation factor-1 α, XM_002284888).

The real time data generated by the Rotor-Gene™ 6000 (Corbett Life Science) analyser was imported into Microsoft Excel and analysed using the GeNorm applet (Vandesompele *et al.* 2002) applying the delta-delta-Ct method (Livak and

Schmittgen 2001). Amplification efficiency coefficients for each individual sample were generated by Rotor-Gene™ 6000 analysis software (Corbett Life Science) and used for data normalisation. For wounding experiments, the data obtained by the delta-delta-Ct method was validated by applying two standard curves methods in the Rotor-Gene 6000 Series software (Corbett Life Science). Standard curves were generated by using serial dilutions of PCR products for each gene. Reference genes used in this study were chosen from the genes that had previously been shown to be appropriate for relative quantification of gene transcripts in grape berries (Reid *et al.* 2006). Calculated transcript levels were normalised against a combination of reference genes using GeNorm (Vandesompele *et al.* 2002).

Heterologous expression of recombinant LOX proteins

Three microlitres of Luria Bertani (LB) medium (Sambrook and Russell 2001) supplemented with ampicillin ($100\text{ }\mu\text{g mL}^{-1}$) was inoculated with a single colony of *E. coli* BL21 (DE3) containing the construct of interest cloned into the *pDEST17* vector as described earlier. A 500- μL aliquot of the overnight culture was inoculated into 500 mL LB with $100\text{ }\mu\text{g mL}^{-1}$ ampicillin culture and grown in an orbital incubator at 250 rpm at 37°C until the optical density of the culture measured at 600 nm (OD_{600}) = 0.6 of recombinant protein expression was induced by addition of isopropyl β -D-1-thiogalactopyranoside (IPTG) to a final concentration of 1 mM. The incubation temperature was then decreased to 14°C and the culture was incubated for a further 18 h at 250 rpm. Cells were harvested by centrifugation at 5000g for 10 min, washed with ice-cold 20 mM sodium phosphate buffer (pH 7.4) and centrifuged again at 5000g. The washed pellet was resuspended in 40 mL of binding-wash buffer (as recommended for use with HisTrap™ FF columns, GE Healthcare Life Sciences, Auckland, New Zealand; 20 mM sodium phosphate, 0.5 M NaCl, 20 mM imidazole, pH 7.4) to which one tablet of Complete EDTA-free Protease Inhibitor Cocktail (Roche, Auckland, New Zealand) was added. The resuspended cells were then lysed using sonication (10 $\times$ 10 s bursts with 10 s breaks on ice at power 7 at 50% power using a SONICATOR W-225; Heat Systems-Ultrasonic Inc., Watson Victor Ltd, Auckland, New Zealand). The resulting slurry was spun at 14 000g for 20 min at 4°C to pellet cell debris.

Purification of recombinant VvLOXA-TP and VvLOXO-TP

The supernatant obtained as described above was filtered through a 45- μm filter and applied to 3 $\times$ 1 mL HisTrap™ FF columns connected in series pre-equilibrated with binding-wash buffer, using a Biologic DuoFlow 10 System peristaltic pump (Bio-Rad, Auckland, New Zealand). The columns were then washed with binding-wash buffer, monitoring absorbance at 280 nm until the absorbance returned to the baseline. Bound protein was then eluted with the elution buffer (20 mM sodium phosphate, 0.5 M NaCl, 500 mM imidazole pH 7.4).

The eluted 6 $\times$ -histidine (His)-purified fraction was exchanged into 30 mM sodium phosphate, 50 mM NaCl (pH 7.5) using 4 $\times$ 5 mL Hi-Trap Desalt columns (GE Healthcare Life Sciences), connected in series. Glycerol was then added to the recovered protein solution to a final

concentration of 30%. Samples were analysed on Nu-PAGE gels (Invitrogen) and the protein concentration was estimated using Quant-iT™ Protein Assay Kit (Invitrogen) on a Qubit fluorometer (Invitrogen). Protein was stored in small aliquots at -80°C until required.

Determination of recombinant LOX activity

Five-millimolar solutions of fatty acid substrates (α -linolenic (LnA), linoleic (LA) or arachadonic (AA) acids) (Sigma Aldrich) were prepared and the assays were carried out as described by Axelrod *et al.* (1981). Formation of fatty acid (FA) hydroperoxides was measured as an increase in absorbance at 234 nm (for LA and LnA) and at 238 nm (for AA) as a result of conjugated double bond formation, using either a T60 UV-Visible spectrophotometer (PG Instruments, Bio-Strategy, Auckland, New Zealand) or FLUOstar Omega microplate reader (BMG Labtech, Alphatec, Auckland, New Zealand). Reaction buffers were prepared with 0.5 pH unit increments as follows: 100 mM sodium citrate (pH 3.0–3.5), 100 mM sodium acetate (pH 4.0–5.5), 100 mM sodium phosphate (pH 6.0–7.5) and 100 mM sodium borate (pH 8.0–9.5).

To estimate the optimum pH for each of the recombinant proteins, the reaction was run across a pH range of 4.0–9.5 with 0.5 unit increments at 25°C. The optimum temperature was estimated by assaying the LOX activity rate at temperatures of 10–50°C at a pH of 5.5 and 7.5 for LOXA-TP and LOXO-TP respectively. A typical assay with LOXA-TP consisted of 60 μM of either LnA, LA or AA substrate, and 2.3 $\mu\text{g mL}^{-1}$ of purified protein; in experiments with LOXO-TP, the final concentrations were 100 μM of FA substrates and 11.6 $\mu\text{g mL}^{-1}$ of protein.

Specific activities for recombinant LOXs were expressed as units per milligram of total protein (U mg^{-1}). One unit for LOXA-TP was calculated as the amount of protein required to produce 1 μmol of 13(S)-hydroperoxyoctadecatrienoic acid (13-HPOTrE) for 1 min at a pH of 5.5 at 25°C. One unit for LOXO-TP was calculated as the amount of protein required to produce 1 μmol of 13-HPOTrE at a pH of 7.5 at 25°C.

Kinetic properties of the recombinant grape LOXs

Kinetic properties of recombinant LOXs were determined by monitoring FA hydroperoxide formation across a range of concentrations of LnA, LA and AA using standard conditions. For LOXA-TP, the standard assay consisted of 0–100 μM of FA substrate in 100 mM sodium acetate buffer (pH 5.5) at 25°C, 2.3 $\mu\text{g mL}^{-1}$ of partially purified recombinant LOXA-TP protein and various final concentrations of Tween 20 (Ajax Finechem, Auckland, New Zealand), depending on the substrate concentration (0.0001% (v/v) of Tween 20 per 1 μM of the substrate). For LOXO-TP, the standard assay consisted of 0–160 μM of FA substrate in a sodium phosphate buffer (pH 7.5) at 25°C, and 0.00003% (v/v) of Tween 20 per 1 μM of the substrate and 11.6 $\mu\text{g mL}^{-1}$ of partially purified recombinant LOXO-TP. Each reaction was run in triplicate. Negative controls (no enzyme and no substrate) were included. The first order kinetic rates were estimated as changes in absorbance at 234 nm s^{-1} and converted into $\mu\text{mol min}^{-1} \text{mg}^{-1}$, using extinction coefficients $\epsilon = 23\,000$ for LnA and LA hydroperoxides and

$\varepsilon = 27\,000$ for AA hydroperoxides. The data was then exported into GraFit software (Erithacus Software Limited, Horley, Surrey, UK) and fitted using a non-linear model and a Lineweaver Burk plot to determine Michaelis constant (K_m) and maximum velocity (V_{max}) values for both enzymes.

HPLC analysis of LOX products

To determine identity of the LOXA-TP and LOXO-TP recombinant protein reaction products, the respective recombinant proteins were incubated with $60\,\mu\text{M}$ of either LnA, LA or AA in 50 mL reactions at 25°C for 15 min, and at pH 5.5 and 7.5 for LOXA-TP and LOXO-TP respectively. Enzyme reactions were stopped by adding 1.6 mL of 2.5 N HCl to adjust the pH of the reaction solution to 3.0. Extract-Clean™ SPE C18 columns (Alltech, Grace Davison Discovery Sciences, Auckland, New Zealand) were used to purify the LOX metabolites according to the manufacturer's instructions. Briefly, the columns were conditioned with methanol and then dionised water. The sample was then placed onto the column, washed with 10% acetonitrile and the samples were eluted with 5 mL of 100% acetonitrile. The volume of the sample was reduced to 1 mL by evaporation of the sample through flushing of the headspace with oxygen-free nitrogen. The concentrated hydroperoxide samples were stored under nitrogen in sealed vials at -80°C before analysis.

To identify LOX products, reverse-phase HPLC was performed on the Agilent 1100 Series HPLC System (Agilent Technologies, Santa Clara, CA, USA), using a Luna $5\,\mu\text{m}$ C18(2) 100 A ($250 \times 4.60\text{ mm}$) column (Phenomenex, Auckland, New Zealand). The products' LOX metabolites and standards were resolved by isocratic elution with acetonitrile + 0.1% formic acid: $\text{H}_2\text{O} + 0.1\%$ formic acid (63:37) at a flow rate of 0.5 mL min^{-1} for 48 min at 25°C . The identity of the LOX products was determined by comparison of the retention times with authentic standards (Sapphire Biosciences, Waterloo, NSW, Australia). The results were further validated by comparison of the absorption spectra of the LOX products with the respective standards.

Results

In silico grape genome mining

To determine the complexity of the grape LOX gene family, the grape genome was mined using sequence analysis and gene prediction software as described in the Materials and methods section. Based on the protein sequence similarity with other functionally characterised plant LOXs, 18 LOX-like sequences of various lengths were identified in the Pinot Noir genome and labelled with letters of the English alphabet starting from *VvLOXA* through to *VvLOXS* in order of their identification (Table 1). Out of the 18 grape LOX sequences identified, 12 encoded full- or nearly full-length functional proteins when compared with LOXs from other species. The remaining six LOX sequences either coded for truncated proteins, or there were unresolvable gaps in the respective genomic sequences, which hampered their *in silico* analysis.

Specifically, analysis of the *VvLOXF* sequence retrieved from Genoscope revealed a two-nucleotide insertion in exon 3, which leads to a shift in the open reading frame and would potentially

result in a premature termination of the encoded protein. However, a homologous sequence of *VvLOXF* obtained from GenBank (XR_078140) lacked such an insertion and encoded fully functional protein. *VvLOXK* possessed two single-nucleotide substitutions within its coding sequence, which resulted in two premature stop-codons being generated. The 5'-region of the *VvLOXM* coding sequence was interrupted by a transposon-like sequence and only 267 bp of the *VvLOXN* coding sequence could be retrieved from the Pinot Noir genome resources. It is likely therefore that these LOXs are putative pseudo-genes. Genomic sequences of *VvLOXD* and *VvLOXR* possessed several unresolvable gaps in the retrievable sequence data, and therefore their full structure and functionality could not be predicted.

The chromosomal location of the identified grape LOXs were placed on the Genoscope predicted grape linkage groups (Table 1, Fig. S1, available as an Accessory publication to this paper). Not all identified LOX sequences, however, could be clearly linked to a particular linkage group or chromosome at this time. Specifically, the sequence of *VvLOXP* was assigned to chromosome 1, but its particular location within the chromosome was not determined and it was therefore placed in a separate linkage group, Chr1_random. Moreover, the sequences of *VvLOXD*, *VvLOXR* and *VvLOXS* were found within a genomic sequence that cannot be associated with any of the currently defined grape chromosomes; therefore, they are located to the ChrUn_random, the collection of unassigned ultracontigs.

Phylogenetic tree construction

Plant lipoxygenases are categorised into three groups based on their catalytic activity (the positional specificity of insertion of the hydroperoxide group) and the presence or absence of an *N*-terminal chloroplast-targeting peptide. Depending on the position of oxygenation on the fatty acid backbone (either at carbon-9 or carbon-13), LOXs can be divided into 9-LOX or 13-LOX subclasses respectively (Liavonchanka and Feussner 2006). 13-LOXs, in turn, can be further split into sub-groups with respect to their protein structure. Type I LOXs lack plastid targeting peptides and share over 75% sequence similarity among themselves, whereas type II LOXs possess a plastid-targeting peptide and exhibit low sequence similarity with members of the sub-group (Shibata and Axelrod 1995). To predict their regiospecificity and potential sub-cellular location, the complete putative amino acid sequences of the identified grape LOXs were aligned to other biochemically characterised plant LOXs and analysed as described in the Materials and methods section. Extensive analysis of the available sequence resources and 5'-RACE experiments failed to recover a full-length *VvLOXD* sequence. However, as the identified *VvLOXD* sequence contained the important LOX functional domains, the available sequence was used in the phylogenetic analysis. *VvLOXM*, *VvLOXN* and *VvLOXR* were predicted to be pseudo-genes with very little sequence information available, and were removed from the alignment and subsequent tree construction.

Most of the predicted grape LOX proteins contained specific polycystin-1, lipoxygenase, alpha-toxin (PLAT) domain and LOX domains and conserved residues required for binding iron. Exceptions were *VvLOXI*, which had an aspartic acid

residue in the place of an histidine, which is critical for binding iron; and *VvLOXS*, which possessed a terminal valine rather than the highly conserved isoleucine typical of plant LOXs (positions 892 and 1058 respectively; Fig. S2).

Phylogenetic analysis placed all identified grape LOXs within either the type II 13-LOXs or 9-LOX groups (Fig. 2). *VvLOXC*, *VvLOXB* and *VvLOXL* were grouped with the characterised 9-LOX members and possessed the conserved valine residues typical for this class of LOXs (position 755; Fig. S2). The remaining LOXs, with the exception of *VvLOXD*, possessed the conserved phenylalanine residue involved in determining the regiospecificity that is typical for 13-LOXs, and were placed in two separate clusters of type II 13-LOXs (position 755; Fig. S2, Fig. 1). *VvLOXD* possessed a leucine residue at this site that was found in only one other LOX, an uncharacterised putative Pt-LOX from Western balsam poplar (*Populus trichocarpa* (Torr. & Gray); position 755, Fig. S2). *VvLOXD* and this Pt-LOX shared 66% amino acid sequence identity and were placed in a distinct branch of the tree (Fig. 2). Further to these

significant differences, *VvLOXD* was the only LOX protein that possessed a serine, rather than an alanine, at the position determining LOX stereospecificity (position 730; Fig. S2).

In order to check whether missing sequence data in the *N*-terminal region of *VvLOXD* in the original alignment affected the grouping and relationships among the other grape LOX members, a second alignment and tree were constructed containing conserved LOX domain regions only. This analysis resulted in the construction of an almost identical tree that did not change the relative position of its members (Fig. S3).

Cloning of *VvLOXA*, -C, -D and -O from Sauvignon Blanc

To initiate functional analysis of grape LOXs in Sauvignon Blanc grape berries, representative members of the LOX groupings determined above were PCR amplified from berry-derived cDNA, cloned into a *pENTR™/TEV/D-TOPO®* vector (Invitrogen) and sequenced as described in the Materials and methods section. *VvLOXA* and *VvLOXO* were selected

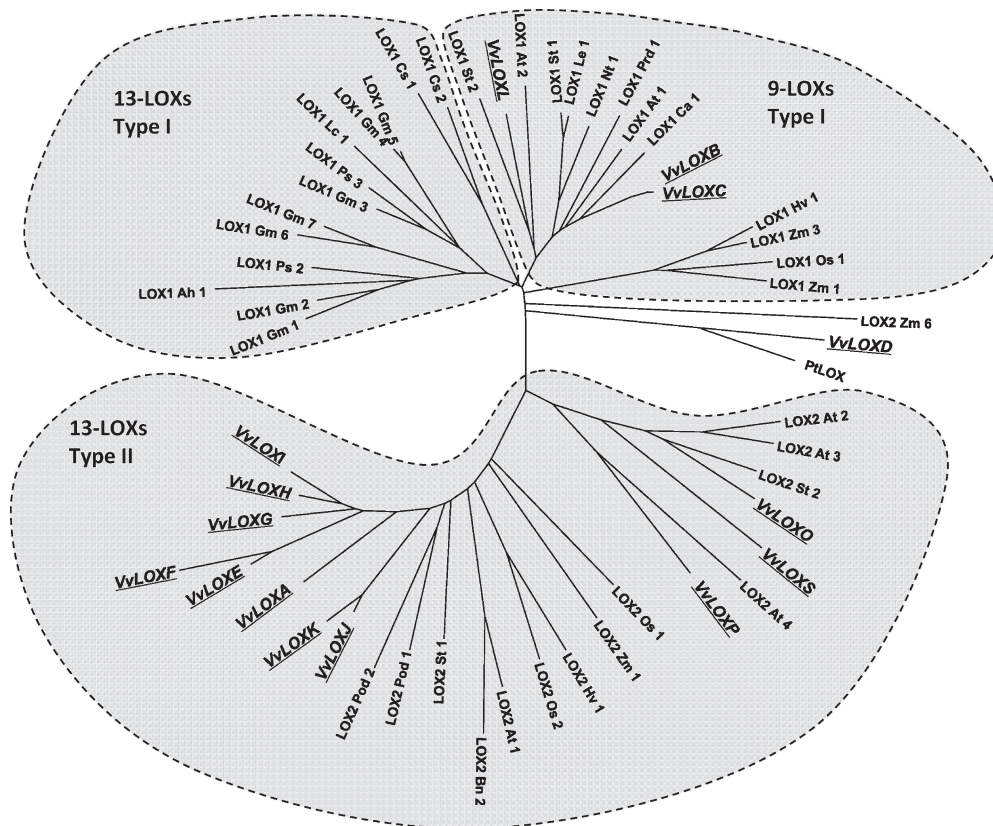


Fig. 2. Phylogenetic analysis of the predicted grape LOXs (underscored) and biochemically characterised plant LOXs. The evolutionary relationships depicted were inferred using the Neighbour-Joining method (Saitou and Nei 1987). The bootstrapped consensus tree was inferred from 1000 replicates (Felsenstein 1985). Those branches corresponding to partitions that were reproduced in less than 50% of the bootstrap replicates were collapsed. The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the Poisson correction method (Zuckerkandl *et al.* 1965) and are in the units of the number of amino acid substitutions per site. All positions containing alignment gaps and missing data were eliminated only in pairwise sequence comparisons. Phylogenetic analyses were conducted in MEGA4 (Tamura *et al.* 2007). The shaded areas outline three separate LOX classes described in the text. Tables 1 and 2 contain gene IDs and accession numbers for the LOXs used in the analysis.

as representatives of the two 13-LOX sub-groupings, while *VvLOXC* and *VvLOXD* were selected as representatives of the 9-LOX and outlying clades respectively. Due to the missing genomic sequence information in the 5' region of *VvLOXD*, we were unable to design PCR primers for the predicted translation initiation site and consequently obtain full-length coding sequences from Sauvignon Blanc. Using 5'-RACE techniques, we obtained a PCR fragment that encoded a putative protein that was 61–180 bp shorter than the nearest BLAST match to *VvLOXD* from other plants.

VvLOXA, *VvLOXC* and *VvLOXD* shared 99.0–99.6% identity in both nucleotide and amino acid sequences with their homologues predicted in the red grape cultivar Pinot Noir. These were submitted to GenBank with the accession numbers FJ858255, FJ858256 and FJ858257 respectively. As only a small region of the *VvLOXD* coding sequence can be retrieved from the Pinot Noir genome at the moment, analysis of the sequence identity for this gene between the two cultivars is currently not meaningful.

The resulting gene sequences were used to generate specific primers for the semiquantitative real-time PCR (qPCR) determination of transcript abundance in Sauvignon Blanc berries. The qPCR analyses were undertaken to study relative expression of LOX genes in grape berries during development, after wounding and upon infection with the common grape pathogen *B. cinerea*.

Distribution of selected Sauvignon Blanc LOX transcripts within berries

As the formation of short-chain aliphatic volatiles are arguably most important in berries just before or at harvest, we were interested in determining the gross patterns of tissue-specific expression of each of the four selected LOXs in berries at later stages of ripeness. We therefore determined the transcript distribution for each LOX in three fractions: berry skin, pulp and seeds. The majority of *VvLOXA* transcripts, ~77% of total *VvLOXA* expression, were located in berry skins, with the remainder of the expression equally distributed between the pulp and seeds (Fig. 3). The opposite was observed for *VvLOXD*, where the majority of the transcript was found to be localised in seeds (75%), followed by skins (21%). Only a small portion of *VvLOXD* transcripts (4%) were detected in the pulp (Fig. 3). Transcripts of *VvLOXC*, the predicted 9-LOX, were found predominantly in pulp (48%), with the remaining expression split equally between the skin and seeds (Fig. 3). *VvLOXD* transcripts were found to be split relatively evenly among all compartments, with the bulk of transcripts located in skin (Fig. 3).

Changes in LOX gene expression over berry development

The changes in LOX expression during berry development was studied during the 2007, 2008 and 2009 growing seasons (Fig. 4a–d). In all three seasons, the *VvLOXA* gene displayed a similar pattern of expression, with an increase in transcript abundance around veraison, which subsequently remained at a relatively constant level through to harvest (Fig. 4a). In contrast, *VvLOXC* transcript abundance steadily declined during the initial stages of berry development and remained low during the post-

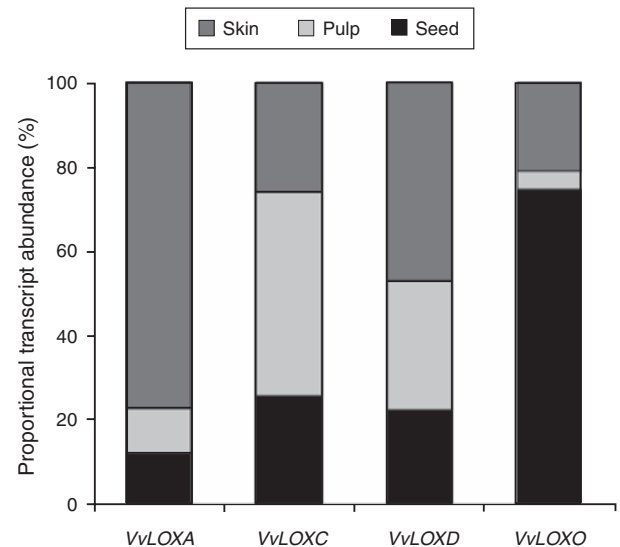


Fig. 3. The distribution of identified LOX gene transcripts was studied in three berry fractions: skin, pulp and seeds. The geometrical mean of expression of the reference genes *Actin*, *GAPDH*, *SAND* and *EF1* was used as a normalisation factor between samples.

veraison stages (Fig. 4b). *VvLOXD* expression varied across development and from season to season (Fig. 4c). *VvLOXD* expression was found to decrease from earlier to later stages in the 2008 and 2009 seasons. Interestingly, in the 2007 season, *VvLOXD* expression stayed low during the earlier stages of development, with an increase beginning in early post-veraison stage and reaching a maximum in the pre-harvest stage (Fig. 4d).

Grape berry LOXs are differentially expressed upon wounding

In grape berries, the four identified LOX genes exhibited varied responsiveness to mechanical damage (Fig. 5a–d).

A clear response to wounding was observed for *VvLOXD*, with rapid accumulation of transcripts observable after 30 min and with peak expression occurring at 3 h after wounding (a ~30-fold increase over basal levels of expression). A rapid decrease back to basal levels was observed 6 h after wounding (Fig. 5d). *VvLOXC* was also marginally increased by wounding (Fig. 5b), reaching maximum levels of expression in 3 h after wounding; after 3 h, expression was maintained above basal levels for up to 24 h. A bimodal pattern of *VvLOXA* expression was also evident, with an initial small decrease in expression followed by a peak at 6 h that was followed by a gradual decline in expression over the remainder of the experiment (Fig. 5a). There were no major changes observed in the expression of *VvLOXD* (Fig. 5c).

Grape berry LOXs are differentially expressed in response to pathogen infection

B. cinerea is a common necrotrophic fungal pathogen infecting grape leaves and fruits, and bringing significant losses in yield and quality of the harvested grapes. It has been previously shown in the white grape cultivar Chardonnay that inoculating leaves with *B. cinerea* can both stimulate and repress pathogen-related genes

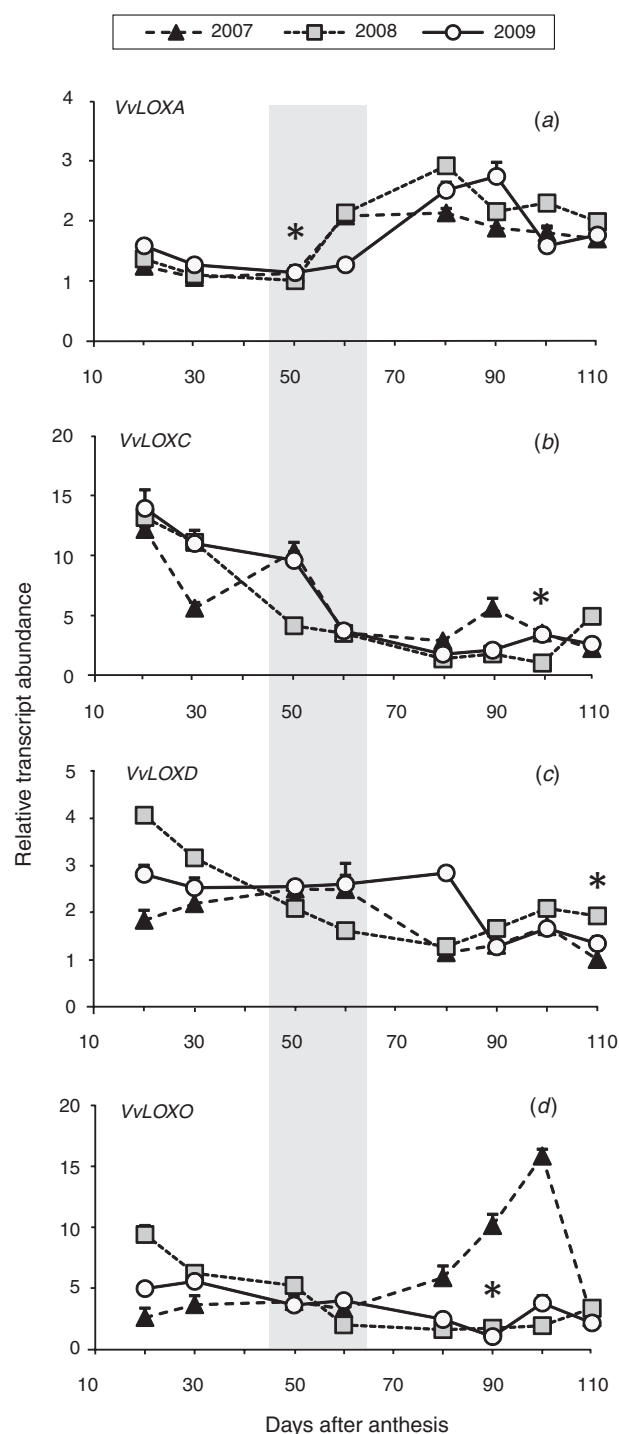


Fig. 4. The relative expression of grape berry LOX genes throughout development was studied in the 2007, 2008 and 2009 growing seasons. Relative transcript abundance levels for (a) *VvLOXA*, (b) *VvLOXC*, (c) *VvLOXD* and (d) *VvLOXO* at days after anthesis are displayed relative to the sample with the lowest level of expression for that experiment (labelled with an asterisk). The geometrical mean of the *Actin* and *GAPDH* reference genes was used as a normalisation factor. The error bars represent a standard deviation of the mean for technical replicates and propagated by normalisation against two reference genes. Veraison in each season took place approximately 50 to 60 days after anthesis (shaded area).

(Chong *et al.* 2008). To determine how grape berry LOXs respond to *B. cinerea* infection, we collected Sauvignon Blanc bunches partially infected with the pathogen and separated infected berries from the berries that had no obvious signs of the infection. Levels of LOX expression in these two groups were compared with the levels of expression in healthy berries, spatially separated from infected berries (Fig. 6a–d).

VvLOXC, *VvLOXD* and *VvLOXO* had increased levels of expression in the infected berries of between 4-fold (*VvLOXD*) and 25-fold (*VvLOXC* and *-O*) compared with the control (Fig. 6b–d). On the contrary, the levels of *VvLOXA* expression declined by almost 3-fold compared with the control (Fig. 6a). No major changes in LOX gene expression was observed in the non-infected berries collected from the same bunches as the infected berries, with the exception of *VvLOXA*, the transcripts of which decreased slightly (NI in Fig. 6a–d).

Biochemical analysis of the 13-LOXs *VvLOXA* and *VvLOXO*

Two putative grape 13-LOXs (*VvLOXA* and *VvLOXO*) were selected for biochemical analyses that were representative of the two sub-groupings of type II 13-LOX sequences retrieved from the grape genome. Both full-length and truncated clones for each LOX, in which predicted targeting and signal peptides were removed, were cloned into the recombinant protein production vector *pDEST17* (see Materials and methods section). Analysis of the *VvLOXA* sequence revealed a 47 amino acid *N*-terminal peptide that is highly likely to target the protein to the chloroplast (data not shown). Analysis of the *VvLOXO* sequence similarly suggested the presence of a 60 amino acid *N*-terminal peptide that may encode sub-cellular targeting information. Consequently, constructs that encoded both full-length and truncated proteins, in which the predicted targeting peptide was removed, were prepared and analysed for their ability to produce functional LOX protein (LOXAFL, LOXOFL, LOXA-TP and LOXO-TP respectively). Only the truncated versions of *VvLOXA* and *VvLOXO* were found to be capable of producing soluble and functional recombinant LOX protein. Recombinant protein produced using the full-length constructs was only recovered in the insoluble fractions of *E. coli* crude extracts. Additionally, there was no detectable activity found for either of the full-length recombinant proteins in the soluble fractions of these same extracts (data not shown).

LOXO-TP recombinant protein was more unstable than LOXA-TP and lost some activities during the purification process. As a consequence, the specific activity of the purified LOXO-TP protein was almost 10-fold lower than the activity of LOXA-TP (Table 3).

Temperature and pH optima

To determine the optimum temperature for both recombinant LOXs, enzyme activity was measured over a range of temperatures (10–50°C) with saturated concentrations (60 μM for LOXA-TP and 100 μM for LOXO-TP) of three common fatty acid substrates (LnA, LA and AA). Maximal activities for both recombinant proteins were observed at 25°C (Table 3, Fig. 7c, d). The activity of LOXA-TP was abolished at 50°C, whereas at 10°C, LOXA-TP retained ~30–50% of its maximum activity,

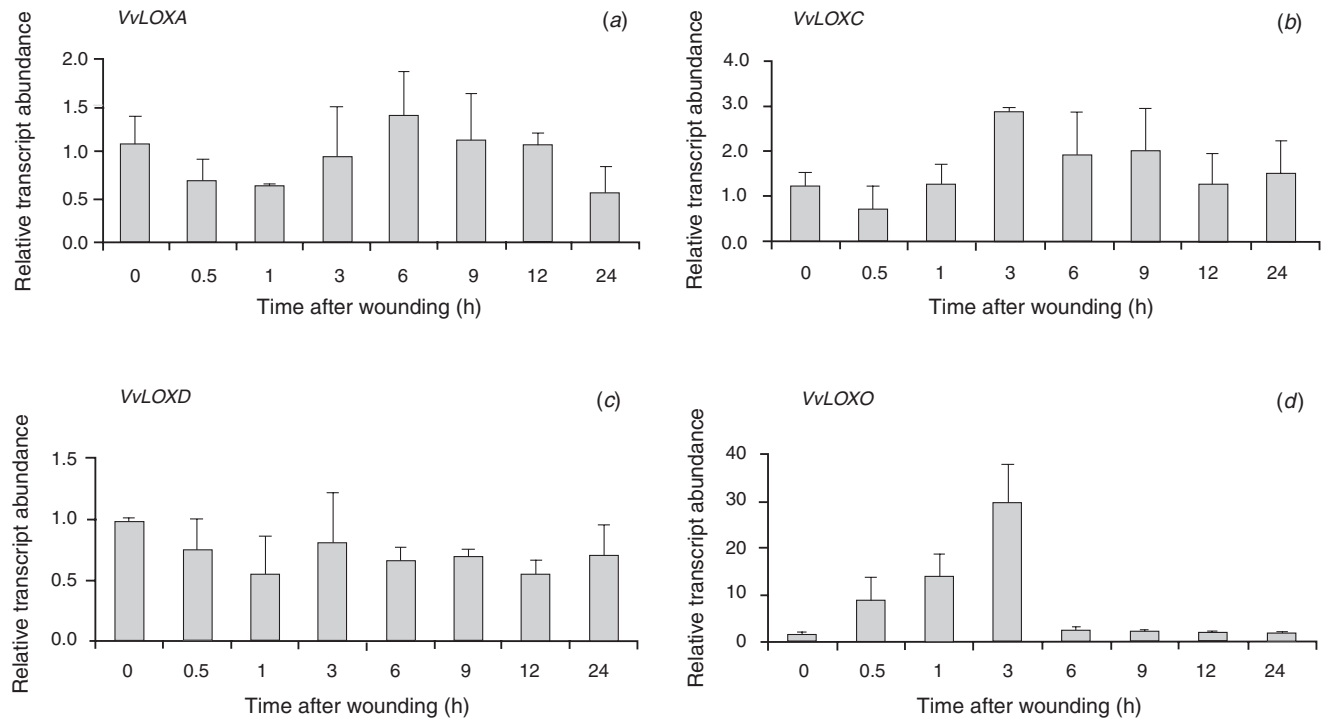


Fig. 5. Expression of grape LOX genes was quantified in mechanically wounded berries. Relative transcript abundance levels for (a) *VvLOXA*, (b) *VvLOXC*, (c) *VvLOXD* and (d) *VvLOXO* were determined at set times after mechanical wounding of grape berries and are displayed relative to the unwounded control (time 0). The geometrical mean of *Actin*, *GAPDH* and *SAND* reference genes was used as a normalisation factor. Three independent biological replicates were analysed for each time point. Each bar represents the mean for the replicated values and the error bars represent standard deviation of the mean.

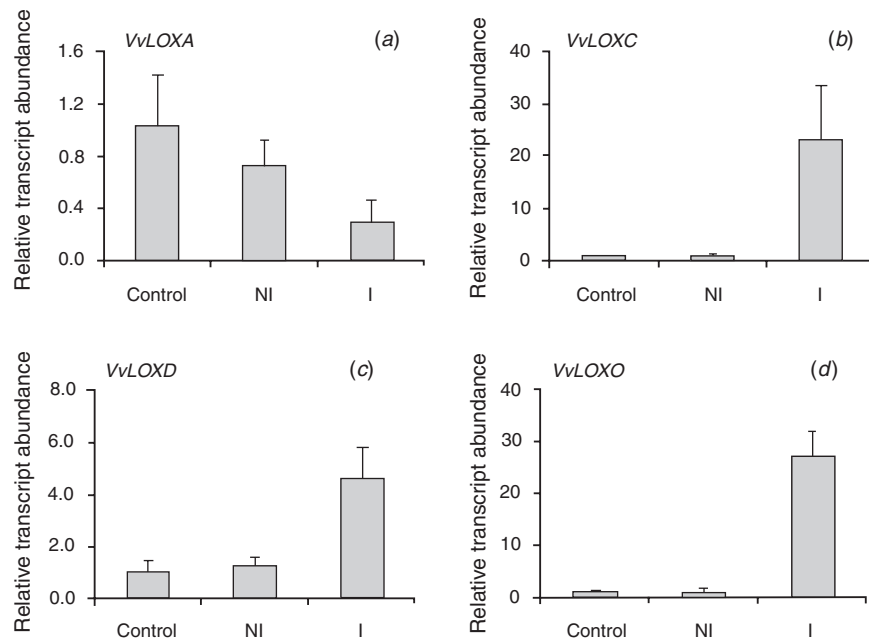


Fig. 6. Effect of pathogen infection on LOX gene expression was studied in grape berries infected with *B. cinerea*. Intact healthy berries were used as a control. Relative transcript abundance of (a) *VvLOXA*, (b) *VvLOXC*, (c) *VvLOXD* and (d) *VvLOXO* was determined in grape berries that were taken from non-infected bunches (Control), berries taken from bunches that did not display symptoms of infection (NI) and berries that showed infection (I). The geometrical mean of *Actin*, *GAPDH* and *SAND* reference genes was used as a normalisation factor. Three independent biological replicates were analysed. Each bar represents a mean for the replicated values and error bars represent standard deviation of the mean.

Table 3. Biochemical properties and optimum assay conditions of recombinant LOXA-TP and LOXO-TP enzymes

Enzyme	Optimum pH	Optimum temperature (°C)	Specific activity (U mg ⁻¹)	Substrate	$V_{\max} \pm \text{s.e.}$ ($\mu\text{mol min}^{-1} \text{mg}^{-1}$)	$K_m \pm \text{s.e.}$ (μM)	$V_{\max} : K_m$	Products formed
LOXA-TP	5.5	25	19.56	LnA	19.56 ± 0.93	3.4 ± 0.6	5.70	13(S)-HPOTrE
				LA	9.29 ± 0.15	1.5 ± 0.1	6.32	13(S)-HPODE
				AA	7.32 ± 0.11	1.0 ± 0.1	7.10	15(S)-HPETE
LOXO-TP	7.5	25	2.14	LnA	2.14 ± 0.04	8.5 ± 0.6	0.25	13(S)-HPOTrE
				LA	0.30 ± 0.02	8.4 ± 1.8	0.04	13(S)-HPODE
				AA	0.05 ± 0.01	8.1 ± 3.1	0.01	15(S)-HPETE

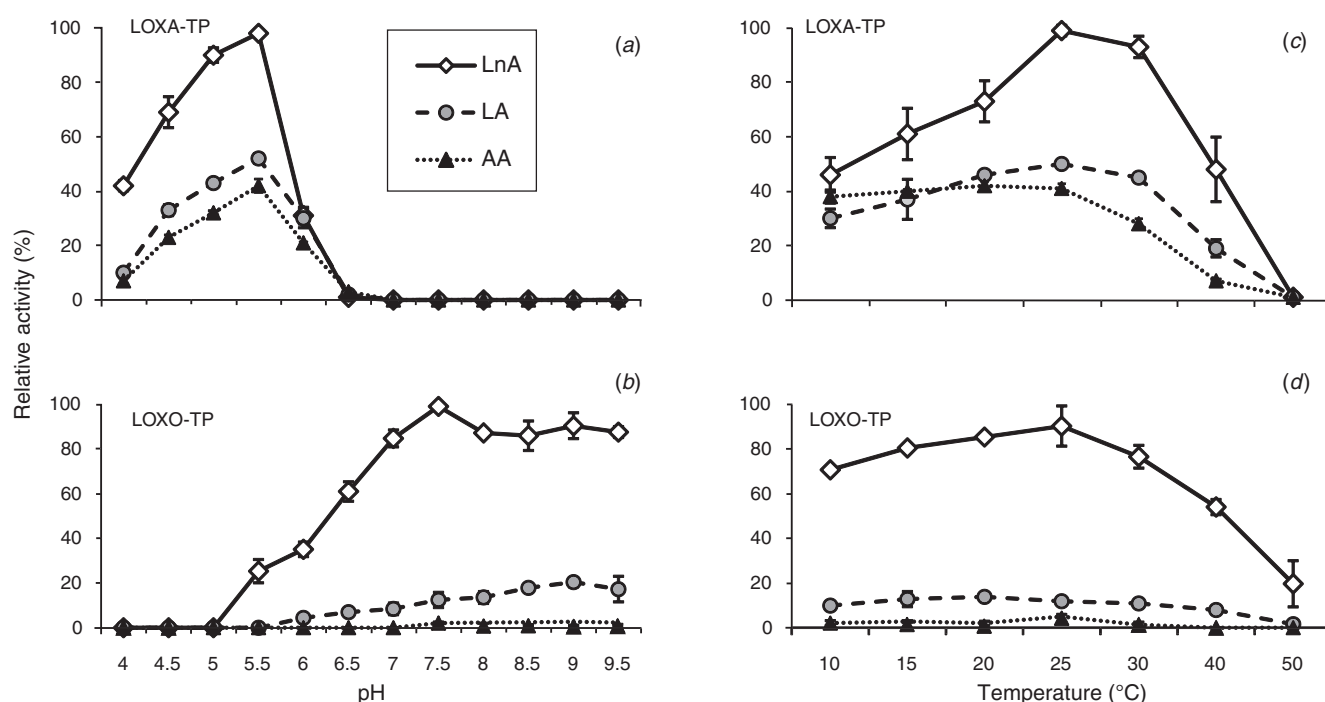


Fig. 7. The effects of pH and temperature on the enzymatic activity of recombinant LOXA-TP (*a*, *c* respectively) and LOXO-TP (*b*, *d* respectively) were determined across a wide range of conditions. LOXA-TP exhibited optimal activity at pH 5.5 (*a*) and a sharp drop in activity on both sides of this pH (*c*). In comparison, LOXO-TP exhibited maximal activity at pH 7.5 (*b*), with little or no activity below pH 5 and significant levels of activity to at least pH 9.5 (*d*). Both recombinant enzymes exhibited maximal activities at 25 °C but differed in the pattern of activity across the temperatures assayed.

depending on the substrate used in the assays (Fig. 7*c*). LOXO-TP was more temperature tolerant and, when incubated with LnA, retained ~20% of its maximal activity at 50°C and retained ~70% of its maximum activity at 10°C (Fig. 7*d*).

To determine the optimal pH for recombinant LOXA-TP and LOXO-TP, LOX assays were conducted in various buffers over a pH range of 3.5–9.5. The assays were performed at 25°C with saturating concentrations of substrate as described above. Precipitation of LOXA-TP occurred at pH 3.5 and LOXO-TP at pH 5.0. Recombinant LOXA-TP protein exhibited its highest activity at pH 5.5 (Table 3, Fig. 7*a*). Utilising LnA as a substrate, LOXA-TP retained ~40% of its maximum activity at pH 4.0 (Fig. 7*a*). LOXA-TP activity dropped by 75% at pH 6.0, only a 0.5 pH unit change from the determined pH optimum of 5.5, and was almost completely deactivated at pH 6.5 with all three FA substrates (Fig. 7*a*). In contrast, LOXO-TP preferred neutral and

alkaline conditions, and had its maximum activities at pH 7.5 with LnA, and pH 9.0 with LA (Table 3, Fig. 7*b*).

Kinetic properties and identification of reaction products

To determine the basic kinetic characteristics of the recombinant LOXA-TP and LOXO-TP proteins, the velocity of the substrate conversion was studied across a range of concentrations of LnA, LA and AA fatty acids using standard conditions as described in the Materials and methods section. The K_m values for both LOXA-TP and LOXO-TP were in the low micromolar range, indicating a very high affinity of both enzymes for all three substrates (Table 3, Fig. S4). A comparison of the K_m and $V_{\max}$ values indicates that LOXA-TP has higher affinity for AA (1.5-fold and 3-fold higher than for LA and LnA respectively), but converts LnA over 2-fold faster than LA and

3-fold faster than AA. In comparison, LOXO-TP had the highest catalytic efficiency with LnA (the $V_{\max} : K_m$ ratios for LnA were 7- and 29-fold higher than for LA and AA respectively), although K_m values for all three fatty acids were very similar (Table 3).

The HPLC analysis of the reaction products produced by LOXA-TP and LOXO-TP revealed that both enzymes convert LnA exclusively into 13(S)-hydroperoxyoctadecatrienoic acid (13(S)-HPOTrE), LA was converted into 13(S)-hydroperoxyoctadecadienoic acid (13(S)-HPODE) and AA into 15(S)-hydroperoxyeicosatetraenoic acid (15(S)-HPETE) (Table 3; Fig. 8). These results confirm the predicted classification of *VvLOXA* and *VvLOXO* as 13-LOX enzymes.

Discussion

Grape LOXs are members of a large gene family

Sequence analysis of the *V. vinifera* genome identified the presence of at least eighteen LOX-like sequences. LOX gene families of a similar size have been reported for potato (*Solanum*

tuberosum L.; at least 14 members (Liavonchanka and Feussner 2006)), rice (*Oryza sativa* L.; at least 16 members (Agrawal *et al.* 2004)), *Medicago truncatula* Gaertn. (15 reported members) and soybean (*Glycine max* (L.) Merr.; at least 19 members (Shin *et al.* 2008)). The expansion of the LOX gene family in legumes has been attributed to ancient polyploidy events with subsequent duplication or tandem duplication of individual loci (Shin *et al.* 2008). Sequencing of the grape genome has provided evidence that the genome of the modern grape has been formed by merging three ancestral genomes either by a true polyploidy event or through successive genome duplications (Jaillon *et al.* 2007). It is therefore quite likely that the size and organisation of the LOX family in grape is the result of similar processes to those observed in legumes.

The size and complexity of the LOX family may prove to be larger than we report here as further data becomes available through improvement of the quality and annotation of the existing reference sequence. Additionally, the highly heterozygous nature and mixed parentage of most grape varieties make it likely that

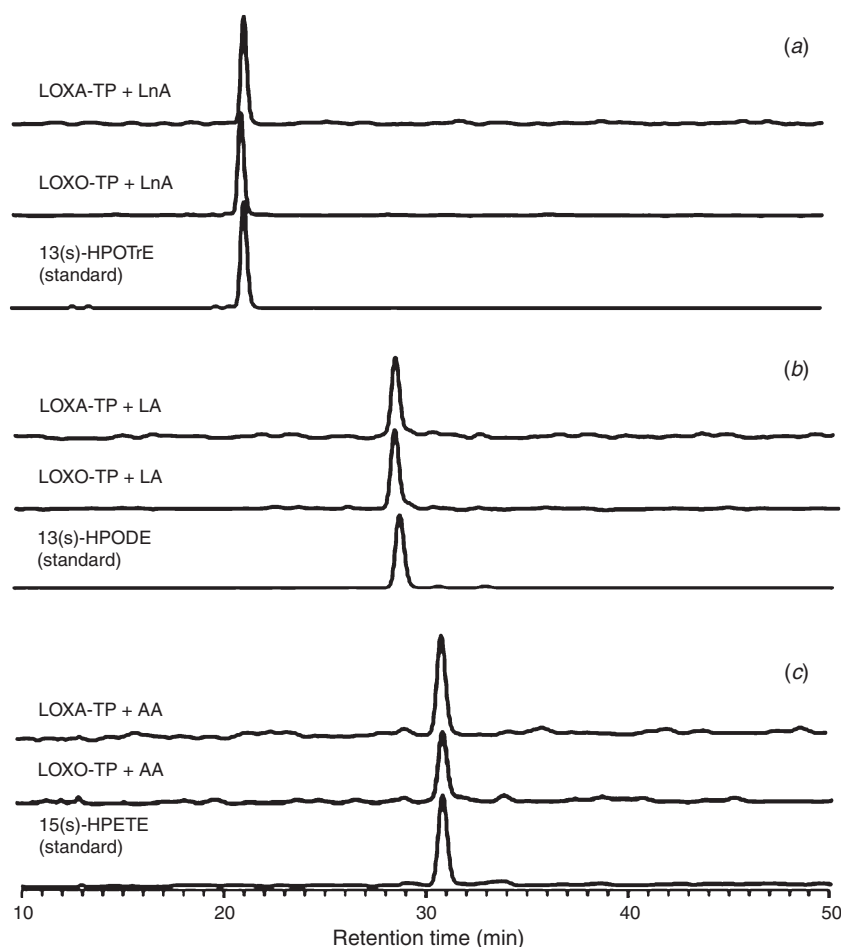


Fig. 8. Product identification of the recombinant LOXA-TP and LOXO-TP. Purified reaction products of LOXA-TP and LOXO-TP activity against LnA, LA and AA were resolved by reverse phase (RP)-HPLC and their retention times were compared with authentic standards 13(S)-HPOTrE, 13(S)-HPODE and 15(S)-HPETE. For each enzyme and with each substrate, the expected 13-hydroperoxides were formed, confirming that both recombinant *VvLOXA* and *VvLOXO* perform as 13-LOXs under standard conditions.

unidentified LOX genes, and differing combinations of functional and non-functional alleles, will be present in *V. vinifera*.

The present classification separates LOX proteins into three groups based on their regiospecificity and the presence or absence of a plastid-targeting peptide. This classification is not perfect, however, and there are several notable exceptions. First, many LOXs possess dual regiospecificity, and *in vitro* experiments have shown that certain LOXs are capable of producing either 9- or 13-hydroperoxides to a greater or lesser extent (Porta and Rocha-Sosa 2002). Second, apart from being localised in the cytoplasm and plastids, LOXs have also been found in a range of cellular organelles, such as the vacuole, peroxysomes, lipid bodies, plasma membranes and microsomal membranes (reviewed in Liavonchanka and Feussner 2006). Furthermore, there are studies indicating that LOXs predicted to be cytosolically localised can be found within the chloroplast (Gao *et al.* 2008). Despite shortcomings, the proposed classification is useful as a first step in predicting potential function and localisation of individual members of large LOX families. It should be noted, however, that functional characterisation of the grape LOX family is required to confirm sub-cellular localisation and function.

As determined by protein structural studies in plant LOXs, five amino acid residues are involved in binding the atom of catalytic iron, namely three conserved histidines and two more conserved amino acids, either histine, asparagine or serine, and the C-terminal isoleucine (Minor *et al.* 1996). *VvLOXI*, *VvLOXR* and *VvLOXS* possessed non-conserved amino acid residues in these conserved regions. *VvLOXI* had a His₆₈₆–Asp substitution (position 892; Fig. S2). The effect of such substitution is unknown, but a single amino acid substitution in one of the iron-binding histidines has been shown to deactivate soybean LOX2 (Wang *et al.* 1994). The predicted sequences of *VvLOXR* and *VvLOXS* both have an apparent substitution of valine for the conserved C-terminal isoleucine (Fig. S2). Interestingly, the same substitution was observed in a unique LOX from maize (*Zea mays* L.) (LOX2_Zm6) that displayed hydroperoxide lyase activity (Gao *et al.* 2008).

Our phylogenetic analysis relates all identified grape LOXs, with the exception of *VvLOXD*, into two groups: type II 13-LOXs and cytosolic 9-LOXs (Fig. 2). The predicted functional grouping of the grape LOXs was supported by the presence of conserved amino acids that determine 13- or 9-LOX specificity. In the region that contains the substrate binding domain, 13-LOXs contain conserved phenylalanine or histidine residues, whereas all 9-LOX members contain valines. The space-filling size of these amino acids has been suggested to affect the substrate orientation or penetration depth inside the binding pocket (Liavonchanka and Feussner 2006). All the predicted grape 13-LOXs contained phenylalanine, while the predicted 9-LOXs contained valine at this position (position 755; Fig. S2). *VvLOXD* had a leucine residue at the regiodetermining position. The impact of such a substitution at this critical position on the regiospecificity of LOX enzymes is unknown, as currently no other functionally characterised LOX has been reported to contain such a substitution.

Most plant LOXs exhibit high product stereospecificity and convert PUFAs almost exclusively into (*S*)-enantiomers, the feature that has been ascribed to a single alanine residue

within the LOX domain (Coffa and Brash 2004). With the exception of *VvLOXD* (which possesses an alanine–serine substitution), all the identified grape LOXs possess a conserved alanine at this site (position 730; Fig. S2). Although no alanine–serine substitution has been functionally characterised to date, the substitution of alanine to glycine in (*S*)-lipoxygenases resulted in the production of (*R*)-enantiomers by these LOXs (Coffa and Brash 2004).

While there are clearly several very interesting substitutions at highly conserved amino acid positions, functional analysis of the individual LOX family members is required to understand the impact, if any, that these substitutions might have on activity. To begin addressing these issues, we focussed initially on analysis of the expression patterns of representative members of the 13-LOX (*VvLOXA* and *VvLOXO*), and 9-LOX (*VvLOXC*) clades, including *VvLOXD* as an outlying member of the wider family. Due to our interest in the role of LOX enzymes in the formation of lipid based aroma and flavour compounds, we further concentrated our efforts on biochemical characterisation of *VvLOXA* and *VvLOXO* as representatives of the subdivision evident within the 13-LOX clade, the class of enzymes proposed to be involved in the formation of C6 metabolites in plants.

Gene expression

Members of the LOX family in plants play a fundamental role in lipid and fatty acid metabolism, which is related to many biological processes and physiological functions. These include the metabolism of storage lipids during germination; turnover of membranes and plastids; formation of signalling molecules such as methyl jasmonate (MeJA), volatile aldehydes and their derivatives upon biotic and abiotic stress; and formation of antimicrobial compounds in response to pathogen attack.

In terms of potential substrates for LOXs localised in grape berries, the most abundant fatty acids of grape berries in both the pericarp and seed are polyunsaturated LA and LnA (Miele *et al.* 1993). The lipid and fatty acid composition of the berry undergoes significant changes across the development, despite relatively little changes in the overall total lipid content (Rouflet *et al.* 1987; Rubio *et al.* 2009). Most of such changes in grape berry pericarp can be attributed to developmentally driven transformation in berry cell membranes, plastids and lipid-like globules (Hardie *et al.* 1996).

Grape berries are a non-climacteric fruit and feature a distinctive stage during the development known as veraison. During veraison, the berries undergo significant changes at a physiological level, which are attributed to berry softening, accumulation of sugars, reduction in acids and changes in colouration in red varieties that signal the onset of ripening (Coombe 1992). These changes are associated with an oxidative burst and changes in cell structure. At this stage, cell division stops and berry enlargement occurs, mainly due to cell expansion. It is sensible, therefore, to assume that the changes in the berry cell structure will lead to changes in lipid metabolism and will subsequently be reflected in changes in the transcription of LOX genes. Furthermore, the importance of LOXs in development can also be deduced from the developmentally-driven evolution of berry oxylipins (Peña-Cortés *et al.* 2004; Kalua and Boss 2009).

All four of the selected LOX genes, *VvLOXA*, *VvLOXC*, *VvLOXD* and *VvLOXO*, were found to be expressed in berries of the white grape cultivar Sauvignon Blanc. These four genes were cloned from this cultivar and, with the exception of *VvLOXD*, full-length clones were obtained. The largest *VvLOXD* fragment obtained by a 5'-RACE experiment contained 2177 bp of protein-coding sequence and was 61–180 bp shorter than its nearest BLAST matches. The comparative studies of LOX transcript distribution within late-stage grape berry fractions indicate that the two 13-LOXs, *VvLOXA* and *VvLOXO*, accumulate in different tissues. In mature berries, *VvLOXA* mRNA transcripts were found predominantly in berry skins (Fig. 3). By contrast, most of the *VvLOXO* basal expression was detected in berry seeds. A previous study of jasmonates in the grape berry reported that concentration of JA and MeJA in seeds is higher than in the skin and pulp; and jasmonates were implicated in influencing seed development and dormancy (Kondo and Fukuda 2001). However, it is yet to be determined whether *VvLOXO* is responsible for the high concentration of jasmonates reported in grape berry seeds. While *VvLOXA* and *-O* were found to be predominantly expressed in the skin and seeds, almost half of the total transcripts of the putative 9-LOX *VvLOXC* were identified in the grape berry pulp (Fig. 3). However, to further our understanding of LOX function within developing berries, this analysis needs to be repeated across all stages of grape berry development to determine whether there are changes in patterns of LOX gene expression within tissues of the developing berry.

Across berry development and across three seasons, *VvLOXA* exhibited a similar pattern of expression, which was characterised by an initial decrease during early stages of development, followed by an increase in expression around veraison (Fig. 4a). Taking into account the high level of *VvLOXA* expression, its predicted chloroplastic location and the fact that most of its transcripts have been found in berry skins of late stage berries, it is appealing to implicate *VvLOXA* involvement in a 'housekeeping' role that is linked to important processes such as plastid turnover and membrane metabolism associated with the switch to berry ripening and subsequent changes in berry structure and metabolism. In contrast to the consistent season-to-season pattern of expression of *VvLOXA*, *VvLOXO* expression during development was both season- and stage-dependent (Fig. 4b). These data may indicate that the *VvLOXO* roles in berry development are not limited to the seed development, as suggested by its transcriptional localisation to seeds of late-stage berries, and that *VvLOXO* expression can be influenced by external cues in seeds or other tissues. The gradual decrease in *VvLOXC* transcripts across berry development (Fig. 4c) may indicate that 9-LOXs are more important during early stages of the development and become less important after veraison.

As mentioned earlier, LOX members can be involved in a plant's response to wounding, most commonly by activating either the allanene oxide synthase (AOS) or hydroperoxide lyase (HPL) branches of the LOX pathway. Generally, the wounding response in plants is often associated with herbivore feeding. In viticulture, this response can also be related to mechanical damage that occurs during fruit or leaf thinning, and during mechanical harvesting and transportation. It has been shown in a range of plants that individual LOX

transcripts rapidly accumulate in response to wounding, which subsequently results in increased levels of volatile jasmonates or GLVs that are released from the wounded plant tissue (Feussner and Wasternack 2002; Matsui 2006).

Wounding resulted in an initial decrease in *VvLOXA* abundance, followed by up to a 2-fold increase; this increase reached its maximum 6 h after wounding (Fig. 5a). A similar response was observed by the nearest *VvLOXA* homologues *AtLOX2* (*LOX2_At_1*) from *Arabidopsis thaliana* (L.) and *LOX-H1* (*LOX2_St_1*) from potato that have been implicated in JA biosynthesis upon wounding (Bell *et al.* 1995; Royo *et al.* 1996). We also observed a 3-fold increase in *VvLOXC* expression within 3 h after wounding and no changes in *VvLOXD* expression at any point after wounding. The most obvious transient response to wounding, however, was observed in *VvLOXO* expression (Fig. 5d). An almost 10-fold increase in *VvLOXO* transcript abundance was observed in 30 min after wounding, reaching a 30-fold peak in expression at 3 h after wounding but returning to basal levels of expression 6 h after wounding. Likewise, the nearest match to *VvLOXO* in potato, *LOX-H3* (*LOX2_St_2*) has shown a rapid accumulation of transcripts in leaves as soon as 30 min after wounding, reducing to almost undetectable levels in 6 h. These results, however, require further investigation to determine which berry tissue (skin, berry or pulp) is the main source of accumulated *VvLOXO* transcripts upon wounding. Interestingly, the early increase in *VvLOXO* transcripts coincided with the initial drop in *VvLOXA* abundance. A similar negative correlation pattern between *VvLOXA* and *VvLOXO* transcript levels was also observed in berries infected with the common grape pathogen *B. cinerea*. While up to a 30-fold increase in *VvLOXO* expression was observed in the berries infected with the pathogen (Fig. 6d), *VvLOXA* transcript levels in the infected berries were 3 times less than in healthy berries (Fig. 6a). These results, together with the expression data obtained from developmental and wounding experiments, may indicate the importance of *VvLOXO* in the response of grape berries to environmental and stress-related stimuli.

Similar to *VvLOXO*, *VvLOXC*, the putative 9-LOX, was also highly expressed in the berries infected with *B. cinerea* (Fig. 6b), although wounding did not seem to influence expression of *VvLOXC* to the same extent (Fig. 5b). Similarly, a putative LOX with partial sequence similarity to *VvLOXC* was tested in leaves of the white grape cultivar Chardonnay inoculated with a spore suspension of *B. cinerea*. This putative LOX showed an increase in transcription 72 h after inoculation (Chong *et al.* 2008). It has been proposed that 9-LOXs are involved in pathogen response by producing anti-microbial compounds or by activating programmed cell death to isolate the site of infection (Göbel *et al.* 2002; Göbel *et al.* 2003). The mechanism, however, of such a response is not completely understood.

Previous studies also indicate that induction of the LOX pathway can be initiated in intact organs and tissues by a systemic response to C6 volatiles and jasmonates released from the damaged tissues (Bate and Rothstein 1998; Yang *et al.* 2009). To determine whether this systemic induction occurs in response to pathogen infection within a bunch, we analysed LOX gene expression in the berries infected with a pathogen, as well as in the non-infected berries adjacent to the

infected berries. Surprisingly, with the exception of a slight change in *VvLOXA* transcription, there was little or no alteration in LOX gene transcription in the non-infected berries compared with the control (bars designated as NI in Fig. 6a–d).

Biochemical characterisation of recombinant *VvLOXA* and *VvLOXO* proteins

As has been mentioned previously, C6 oxylipins, which are abundant in grape juice and wine, have been implicated as important flavour and aroma precursors (Ramey *et al.* 1986; Schneider *et al.* 2006). *VvLOXA* and *VvLOXO* were chosen as the representatives for the two sub-groups of the 13-LOX clade. As 13-LOXs, they are likely to be involved in the formation of C6 compounds and, consequently, they have been expressed as recombinant proteins and assayed *in vitro*.

Expression of the full-length proteins for *VvLOXA* and *VvLOXO* failed to produce any soluble protein, most probably due to the presence of targeting peptides that may have inappropriately targeted the proteins to *E. coli* membrane fractions or, more likely, directed the recombinant protein into *E. coli* inclusion bodies. In contrast, the constructs encoding recombinant *VvLOXA* and *VvLOXO* proteins, in which the targeting peptide had been removed (LOXA-TP and LOXO-TP respectively), successfully produced soluble proteins that exhibited LOX activity.

Two major PUFAs in plants are LnA (18:3) and LA (18:2). However, in our experiments, we also checked the activity of the recombinant grape LOXs against AA (20:4). Although it is absent in plants, AA is commonly found in microorganisms, including common plant pathogens. It has been shown that during pathogen infection, plant LOXs can metabolise a pathogen's AA, thereby activating the plant's defence mechanism (Bostock *et al.* 1992). As was predicted, both recombinant proteins possessed 13-LOX activity, producing 13(S)-HPOTrE, 13(S)-HPODE and 15(S)-HPETE by metabolising LnA, LA and AA fatty acids respectively (Table 3, Fig. 8). Although producing the same products, LOXA-TP and LOXO-TP exhibited significantly different kinetic properties and preferred different *in vitro* reaction conditions for maximal activity (Table 3, Fig. S4). In contrast to LOXO-TP, which preferred alkaline to neutral assay conditions, LOXA-TP was the most active at pH 5.5 and completely inactive at pH 6.5. Moreover, the ability of LOXA-TP to retain its activity at a pH as low as 4 indicates that *VvLOXA* could be responsible for the formation of C6 compounds at the acidic conditions found in grape must.

At saturating substrate concentrations, both LOXA-TP and LOXO-TP recombinant enzymes exhibited a preference for LnA (Table 3, Fig. 7). Moreover, the ability of recombinant LOXO-TP to metabolise LA and AA was very poor in all conditions (Fig. 7b, d). In comparison with LnA, LOXA-TP had more than 2-fold higher catalytic rates at its optimum temperature of 25°C, but had similar catalytic rates at low temperatures with all three substrates (Fig. 7c). Although both enzymes displayed the highest catalytic rates at 25°C, recombinant LOXO-TP protein was more tolerant to high temperatures, retaining around 20% of its maximum activity at 50°C; at this temperature, recombinant LOXA-TP protein was inactive (Fig. 7c, d). A similar effect of

temperature has also been observed in olives (*Olea europaea* L.), where temperatures above 25°C during malaxation resulted in a decline in both LOX and HPL activity, and subsequently resulted in low C6 volatile content released from milled olive fruits (Sánchez and Harwood 2002).

It is evident that the two 13-LOXs are catalytically active and possess the predicted 13-LOX activity. The analysis of the biochemical activity suggests that, *in vivo*, *VvLOXA* and *VvLOXO* are likely to reside in different intracellular environments. The presence of putative targeting peptides and *in silico* analysis of these sequences suggests that both enzymes may be localised in plastids or chloroplasts. The optimal pH for the recombinant LOXA-TP (5.5; Fig. 7a) suggests that a native *VvLOXA* enzyme may reside in a low pH compartment of the chloroplast such as the thylakoid lumen, which can reach pH levels lower than pH 6.0 (Kramer *et al.* 1999). The higher optimal pH observed for recombinant LOXO-TP protein (Fig. 7b) suggests that its native form may be localised in a different compartment. This might be a separate compartment within the chloroplast but outside of the chloroplast lumen, or *VvLOXO* may co-locate with *VvLOXA*, in which case it is unlikely to be active during the day when lumen pH levels are proposed to fall to low levels.

We are currently pursuing further investigation of the identified LOXs in order to ascertain differences in biochemical activity, sub-cellular localisation and function *in vivo*. We are also extending our scope to study LOXs in other grape organs such as leaves, shoots, roots and flowers. Such studies will help to understand the involvement of LOXs in grape physiology as well as contribute to our understanding of the processes that influence the aroma and flavour composition of grape-derived beverages such as wine and grape juice.

Acknowledgements

The authors thank Dr Mike Trought for his invaluable intellectual input into the viticultural aspects of this study. We also thank Pernod Ricard, New Zealand, for their generous use of commercial vines within the Booker vineyard on the Brancott estate for our field materials, and Marlborough Wine Research staff for help in sample collecting. This work was funded by the New Zealand Foundation for Research, Science and Technology, contract number UOAX0404.

References

- Agrawal GK, Tamogami S, Han O, Iwahashi H, Rakwal R (2004) Rice octadecanoid pathway. *Biochemical and Biophysical Research Communications* **317**(1), 1–15. doi:10.1016/j.bbrc.2004.03.020
- Axelrod B, Cheesborough TM, Laakso S (1981) Lipoxygenase from soybean. *Methods in Enzymology* **71**, 441–451. doi:10.1016/0076-6879(81)71055-3
- Bate NJ, Rothstein SJ (1998) C6-volatiles derived from the lipoxygenase pathway induce a subset of defense-related genes. *The Plant Journal* **16**(5), 561–569. doi:10.1046/j.1365-3113x.1998.00324.x
- Baumes R, Bayonove C, Barillère JM, Escudier JL, Cordonnier R (1988) La macération pelliculaire dans la vinification en blanc. Incidence sur la composante volatile des moûts. *Connaissance de la Vigne et du Vin* **32**(3), 209–223.
- Bell E, Creelman RA, Mullet JE (1995) A chloroplast lipoxygenase is required for wound-induced jasmonic acid accumulation in *Arabidopsis*. *Proceedings of the National Academy of Sciences of the United States of America* **92**(19), 8675–8679. doi:10.1073/pnas.92.19.8675

- Bostock RM, Yamamoto H, Choi D, Ricker KE, Ward BL (1992) Rapid stimulation of 5-lipoxygenase activity in potato by the fungal elicitor arachidonic acid. *Plant Physiology* **100**(3), 1448–1456. doi:10.1104/pp.100.3.1448
- Casey R, West SI, Hardy D, Robinson DS, Wu Z, Hughes RK (1999) New frontiers in food enzymology: recombinant lipoxygenases. *Trends in Food Science & Technology* **10**(9), 297–302. doi:10.1016/S0924-2244(00)00010-8
- Chkaiban L, Botondi R, Bellincontro A, De Santis D, Kefalas P, Mencarelli F (2007) Influence of postharvest water stress on lipoxygenase and alcohol dehydrogenase activities, and on the composition of some volatile compounds of Gewürztraminer grapes dehydrated under controlled and uncontrolled thermohygrometric conditions. *Australian Journal of Grape and Wine Research* **13**(3), 142–149. doi:10.1111/j.1755-0238.2007.tb00244.x
- Chong J, Le Henanff G, Bertsch C, Walter B (2008) Identification, expression analysis and characterization of defense and signaling genes in *Vitis vinifera*. *Plant Physiology and Biochemistry* **46**(4), 469–481. doi:10.1016/j.plaphy.2007.09.010
- Clarke RJ, Bakker J (2004) 'Wine flavour chemistry.' (Blackwell Publishing: Oxford, UK)
- Coffa G, Brash AR (2004) A single active site residue directs oxygenation stereospecificity in lipoxygenases: stereocontrol is linked to the position of oxygenation. *Proceedings of the National Academy of Sciences of the United States of America* **101**(44), 15579–15584. doi:10.1073/pnas.0406727101
- Coombe BG (1992) Research on development and ripening of the grape berry. *American Journal of Enology and Viticulture* **43**(1), 101–110.
- da Silva FG, Iandolino A, Al-Kayal F, Bohlmann MC, Cushman MA, et al. (2005) Characterizing the grape transcriptome. Analysis of expressed sequence tags from multiple *Vitis* species and development of a compendium of gene expression during berry development. *Plant Physiology* **139**(2), 574–597. doi:10.1104/pp.105.065748
- Emanuelsson O, Nielsen H, von Heijne G (1999) ChloroP, a neural network-based method for predicting chloroplast transit peptides and their cleavage sites. *Protein Science* **8**(5), 978–984. doi:10.1110/ps.8.5.978
- Emanuelsson O, Brunak S, von Heijne G, Nielsen H (2007) Locating proteins in the cell using TargetP, SignalP and related tools. *Nature Protocols* **2**(4), 953–971. doi:10.1038/nprot.2007.131
- Felsenstein J (1985) Confidence limits on phylogenies: An approach using the bootstrap. *Evolution* **39**, 783–791.
- Feussner I, Wasternack C (2002) The lipoxygenase pathway. *Annual Review of Plant Biology* **53**, 275–297. doi:10.1146/annurev.arplant.53.100301.135248
- Gao X, Stumpe M, Feussner I, Kolomiets M (2008) A novel plastidial lipoxygenase of maize (*Zea mays*) ZmLOX6 encodes for a fatty acid hydroperoxide lyase and is uniquely regulated by phytohormones and pathogen infection. *Planta* **227**(2), 491–503. doi:10.1007/s00425-007-0634-8
- Göbel C, Feussner I, Hamberg M, Rosahl S (2002) Oxylin profiling in pathogen-infected potato leaves. *Biochimica et Biophysica Acta (BBA) – Molecular and Cell Biology of Lipids* **1584**(1), 55–64. doi:10.1016/S1388-1981(02)00268-8
- Göbel C, Feussner I, Rosahl S (2003) Lipid peroxidation during the hypersensitive response in potato in the absence of 9-lipoxygenases. *The Journal of Biological Chemistry* **278**(52), 52834–52840. doi:10.1074/jbc.M310833200
- Hardie WJ, Aggenbach SJ, Jaudzems VG (1996) The plastids of the grape pericarp and their significance in isoprenoid synthesis. *Australian Journal of Grape and Wine Research* **2**(3), 144–154. doi:10.1111/j.1755-0238.1996.tb00102.x
- Horton P, Park K-J, Obayashi T, Fujita N, Harada H, et al. (2007) WoLF PSORT: protein localization predictor. *Nucleic Acids Research* **35**(Suppl. 2), W585–W587. doi:10.1093/nar/gkm259
- Jaillon O, Aury JM, Noel B, Policriti A, Clepet C, et al. (2007) The grapevine genome sequence suggests ancestral hexaploidization in major angiosperm phyla. *Nature* **449**(7161), 463–467. doi:10.1038/nature06148
- Kalua CM, Boss PK (2009) Evolution of volatile compounds during the development of Cabernet Sauvignon grapes (*Vitis vinifera* L.). *Journal of Agricultural and Food Chemistry* **57**(9), 3818–3830. doi:10.1021/jf803471n
- Kondo S, Fukuda K (2001) Changes of jasmonates in grape berries and their possible roles in fruit development. *Scientia Horticulturae* **91**(3–4), 275–288. doi:10.1016/S0304-4238(01)00271-0
- Kramer DM, Sacksteder CA, Cruz JA (1999) How acidic is the lumen? *Photosynthesis Research* **60**(2–3), 151–163. doi:10.1023/A:1006212014787
- Liavonchanka A, Feussner I (2006) Lipoxygenases: occurrence, functions and catalysis. *Journal of Plant Physiology* **163**(3), 348–357. doi:10.1016/j.jplph.2005.11.006
- Livak KJ, Schmittgen TD (2001) Analysis of relative gene expression data using real-time quantitative PCR and the 2^{-ΔΔCT} method. *Methods (San Diego, Calif.)* **25**(4), 402–408. doi:10.1006/meth.2001.1262
- Lund S, Peng F, Nayar T, Reid K, Schlosser J (2008) Gene expression analyses in individual grape (*Vitis vinifera* L.) berries during ripening initiation reveal that pigmentation intensity is a valid indicator of developmental staging within the cluster. *Plant Molecular Biology* **68**(3), 301–315. doi:10.1007/s11103-008-9371-z
- Lund CM, Thompson MK, Benkwitz F, Wohler MW, Triggs CM, et al. (2009) New Zealand Sauvignon Blanc distinct flavor characteristics: sensory, chemical, and consumer aspects. *American Journal of Enology and Viticulture* **60**(1), 1–12.
- Matsui K (2006) Green leaf volatiles: hydroperoxide lyase pathway of oxylipin metabolism. *Current Opinion in Plant Biology* **9**(3), 274–280. doi:10.1016/j.pbi.2006.03.002
- Miele A, Bouard J, Bertrand A (1993) Fatty acids from lipid fractions of leaves and different tissues of Cabernet Sauvignon grapes. *American Journal of Enology and Viticulture* **44**(2), 180–186.
- Minor W, Steczko J, Stec B, Otwinowski Z, Bolin JT, et al. (1996) Crystal structure of soybean lipoxygenase L-1 at 1.4 Å resolution. *Biochemistry* **35**(33), 10687–10701. doi:10.1021/bi960576u
- Peña-Cortés H, Barrios P, Dorta F, Polanco V, Sánchez C, et al. (2004) Involvement of jasmonic acid and derivatives in plant responses to pathogens and insects and in fruit ripening. *Journal of Plant Growth Regulation* **23**(3), 246–260.
- Petit A-N, Wojnarowicz G, Panon M-L, Baillieu F, Clément C, et al. (2009) Botryticides affect grapevine leaf photosynthesis without inducing defense mechanisms. *Planta* **229**(3), 497–506. doi:10.1007/s00425-008-0849-3
- Porta H, Rocha-Sosa M (2002) Plant lipoxygenases. Physiological and molecular features. *Plant Physiology* **130**(1), 15–21. doi:10.1104/pp.010787
- Ramey D, Bertrand A, Ough CS, Singleton VL, Sanders E (1986) Effects of skin contact temperature on Chardonnay must and wine composition. *American Journal of Enology and Viticulture* **37**(2), 99–106.
- Reid K, Olsson N, Schlosser J, Peng F, Lund S (2006) An optimized grapevine RNA isolation procedure and statistical determination of reference genes for real-time RT-PCR during berry development. *BMC Plant Biology* **6**(1), 27. doi:10.1186/1471-2229-6-27
- Roufet M, Bayonove C, Cordonnier R (1987) Etude de la composition lipidique du raisin, *Vitis vinifera* L.: évolution au cours de la maturation et localisation dans la baie [Lipid composition of grapevine berries, *Vitis vinifera* L.: changes during maturation and localization in the berry]. *Vitis* **26**, 85–97.

- Royo J, Vancanneyt G, Pérez AG, Sanz C, Störmann K, *et al.* (1996) Characterization of three potato lipoxygenases with distinct enzymatic activities and different organ-specific and wound-regulated expression patterns. *The Journal of Biological Chemistry* **271**(35), 21012–21019. doi:[10.1074/jbc.271.35.21012](https://doi.org/10.1074/jbc.271.35.21012)
- Rubio M, Alvarez-Ortí M, Alvarruiz A, Fernández E, Pardo JE (2009) Characterization of oil obtained from grape seeds collected during berry development. *Journal of Agricultural and Food Chemistry* **57**(7), 2812–2815. doi:[10.1021/jf803627t](https://doi.org/10.1021/jf803627t)
- Saitou N, Nei M (1987) The Neighbor-Joining method: a new method for reconstructing phylogenetic trees. *Molecular Biology and Evolution* **4**(4), 406–425.
- Sambrook J, Russell DW (Eds) (2001) 'Molecular cloning: a laboratory manual.' 3rd edn. (Cold Spring Harbor Laboratory Press: Cold Spring Harbor, New York)
- Sánchez J, Harwood JL (2002) Biosynthesis of triacylglycerols and volatiles in olives. *European Journal of Lipid Science and Technology* **104**(9–10), 564–573. doi:[10.1002/1438-9312\(200210\)104:9/10<564::AID-EJLT564>3.0.CO;2-5](https://doi.org/10.1002/1438-9312(200210)104:9/10<564::AID-EJLT564>3.0.CO;2-5)
- Schneider R, Charrier F, Razungles A, Baumes R (2006) Evidence for an alternative biogenetic pathway leading to 3-mercaptohexanol and 4-mercapto-4-methylpentan-2-one in wines. *Analytica Chimica Acta* **563**(1–2), 58–64. doi:[10.1016/j.aca.2006.01.057](https://doi.org/10.1016/j.aca.2006.01.057)
- Schwab W, Davidovich-Rikanati R, Lewinsohn E (2008) Biosynthesis of plant-derived flavor compounds. *The Plant Journal* **54**(4), 712–732. doi:[10.1111/j.1365-3113X.2008.03446.x](https://doi.org/10.1111/j.1365-3113X.2008.03446.x)
- Shibata D, Axelrod B (1995) Plant lipoxygenases. *Journal of Lipid Mediators and Cell Signalling* **12**(2–3), 213–228. doi:[10.1016/0929-7855\(95\)00020-Q](https://doi.org/10.1016/0929-7855(95)00020-Q)
- Shibata D, Slusarenko A, Casey R, Hildebrand D, Bell E (1994) Lipoxygenases. *Plant Molecular Biology Reporter* **12**, S41–S42. doi:[10.1007/BF02671567](https://doi.org/10.1007/BF02671567)
- Shin JH, Van K, Kim DH, Kim KD, Jang YE, *et al.* (2008) The lipoxygenase gene family: a genomic fossil of shared polyploidy between *Glycine max* and *Medicago truncatula*. *BMC Plant Biology* **8**(1), 133. doi:[10.1186/1471-2229-8-133](https://doi.org/10.1186/1471-2229-8-133)
- Stumpe M, Feussner I (2006) Formation of oxylipins by CYP74 enzymes. *Phytochemistry Reviews* **5**(2–3), 347–357. doi:[10.1007/s11101-006-9038-9](https://doi.org/10.1007/s11101-006-9038-9)
- Tamura K, Dudley J, Nei M, Kumar S (2007) MEGA4: molecular evolutionary genetics analysis (MEGA) software version 4.0. *Molecular Biology and Evolution* **24**(8), 1596–1599. doi:[10.1093/molbev/msm092](https://doi.org/10.1093/molbev/msm092)
- Thompson JD, Higgins DG, Gibson TJ (1994) CLUSTAL W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. *Nucleic Acids Research* **22**(22), 4673–4680. doi:[10.1093/nar/22.22.4673](https://doi.org/10.1093/nar/22.22.4673)
- Vandesompele J, De Preter K, Pattyn F, Poppe B, Van Roy N, *et al.* (2002) Accurate normalization of real-time quantitative RT-PCR data by geometric averaging of multiple internal control genes. *Genome Biology* **3**(7), research0034.0031–research0034.0011. doi:[10.1186/gb-2002-3-7-research0034](https://doi.org/10.1186/gb-2002-3-7-research0034)
- Wang WH, Takano T, Shibata D, Kitamura K, Takeda G (1994) Molecular basis of a null mutation in soybean lipoxygenase 2: substitution of glutamine for an iron-ligand histidine. *Proceedings of the National Academy of Sciences of the United States of America* **91**(13), 5828–5832. doi:[10.1073/pnas.91.13.5828](https://doi.org/10.1073/pnas.91.13.5828)
- Wasternack C (2007) Jasmonates: an update on biosynthesis, signal transduction and action in plant stress response, growth and development. *Annals of Botany* **100**(4), 681–697. doi:[10.1093/aob/mcm079](https://doi.org/10.1093/aob/mcm079)
- Yang H-R, Tang K, Liu H-T, Pan Q-H, Huang W-D (2009) Jasmonic acid is induced in a biphasic manner in response of pea seedlings to wounding. *Journal of Integrative Plant Biology* **51**(6), 562–573. doi:[10.1111/j.1744-7909.2009.00832.x](https://doi.org/10.1111/j.1744-7909.2009.00832.x)
- Zuckerkindl E, Pauling L, Bryson V, Vogel HJ (1965) Evolutionary divergence and convergence in proteins. In 'Evolving genes and proteins'. (Eds V Bryson, HJ Vogel) pp. 97–166. (Academic Press: New York)

Manuscript received 9 November 2009, accepted 12 April 2010